TH650A/TS2000

Industrial Robot

INSTRUCTION MANUAL

INDUSTRIAL ROBOT

TRANSPORTATION AND INSTALLATION MANUAL

Notice

1. Make sure that this instruction manual is delivered to the final user of SHIBAURA MACHINE's industrial robot.
2. Before operating the industrial robot, read through and completely understand this manual.
3. After reading through this manual, keep it nearby for future reference.

Rev.4

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NUMAZU, JAPAN

Shibaura Machine

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Preface

This manual describes the basic specifications of the industrial robot and controller, and how to unpack and install them. Specifically, it describes how to unpack the shipment containing the equipment, how to install the equipment, how to connect wiring and piping, and how to attach tools. Be sure to look through this manual before unpacking the shipment.

Before beginning the work according to this manual, read through the Safety Manual so that you can understand the safety measures.

This manual is divided into the following five (5) sections:

Section 1 Specifications
This section describes the basic specifications and names of respective units of the robot and controller.

Section 2 Transportation
This section describes how to remove the robot and controller from their boxes and how to transport them to the installation site. This section also discusses how to temporarily store the equipment after unpacking the shipment.

Section 3 Installation
This section discusses the equipment installation environment, space requirements, and how to install the equipment.

Section 4 System Connections
This section describes how to connect the robot, controller and peripheral equipment.

Section 5 Tool Interface
This section describes how to connect the tool to the robot arm and how to connect pipes and wires to the tool. This section also discusses maximum permissible loads of the tool.

Precautions on Safety

Important information on the robot and controller is noted in the instruction manual to prevent injury to the user and persons nearby, prevent damage to assets and to ensure correct use.
Make sure that the following details (indications and symbols) are well understood before reading this manual. Always observe the information that is noted.

[Explanation of indications]

Indication	Meaning of indication
! **DANGER**	This means that "incorrect handling will lead to fatalities or major injuries".
! **WARNING**	This means that "incorrect handling will lead to fatalities or serious injuries."
! **CAUTION**	This means that "incorrect handling may lead to personal injuries *1) or physical damage *2)".

*1) Injuries refer to injuries, burns and electric shocks, etc., which do not require hospitalization or long term treatment.

*2) Physical damage refers to damages due to destruction of assets or resources.

[Explanation of symbols]

Symbol	Meaning of symbol
	This means that the action is prohibited (must not be done). The details of the actions actually prohibited are indicated with pictures or words in or near the symbol.
	This means that the action is mandatory (must be done). The details of the actions that must be done are indicated with pictures or words in or near the symbol.
	This means danger. The details of the actual danger are indicated with pictures or words in or near the symbol.
	This means caution. The details of the actual caution are indicated with pictures or words in or near the symbol.

CAUTION

- Always read through the Safety Manual provided separately before starting actual work to ensure safety work covering from the robot installation to operation.

[Installation and transportation]

Always observe the following items to safely use the robot.

DANGER	
 Prohibited	• DO NOT install or operate if any parts are damaged or missing. Doing so could lead to electric shocks, fires or faults. • DO NOT install the robot where it may be subject to fluids such as water. Doing so could lead to electric shocks, fires or faults. • Do not place the robot near combustible matters. Doing so could lead to fires if the matter ignites due to a fault, etc.
 Mandatory	• Always secure the robot with the attached clamps before transporting it. Failure to do so could lead to injuries if the arm moves when the robot is suspended. • Wire the robot after installation. Wiring the robot before installation could lead to electric shocks or injuries. • Always use the power voltage and power capacity designated by SHIBAURA MACHINE. Failure to do so could lead to device faults or fires. • Always use the designated power cable. Using a cable other than that designated could lead to fires or faults.
 Always ground	• Completely connect the grounding cable. Failure to do so could lead to electric shocks or fires if a fault or fault current occurs. Noise could lead to malfunction. Also, it could cause mis-operation by noise.

! CAUTION	
 Prohibited	• NEVER lift the robot by the arm 2 cover or arm 2. Doing so will apply an excessive force on the robot's mechanism section and could lead to faults. • For the controller, secure the ample space for air vent. Heating of controller could lead to malfunction.
 Mandatory	• When lifting the robot, lift it up slowly. The robot will tilt slightly, so lifting it up suddenly could be hazardous. • When storing the robot, secure it to the base. The robot will be unstable if just set down, and it could tilt over.

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1. Specifications

1.1 Name of Each Part

Fig. 1.1 shows the name of each part of the robot.

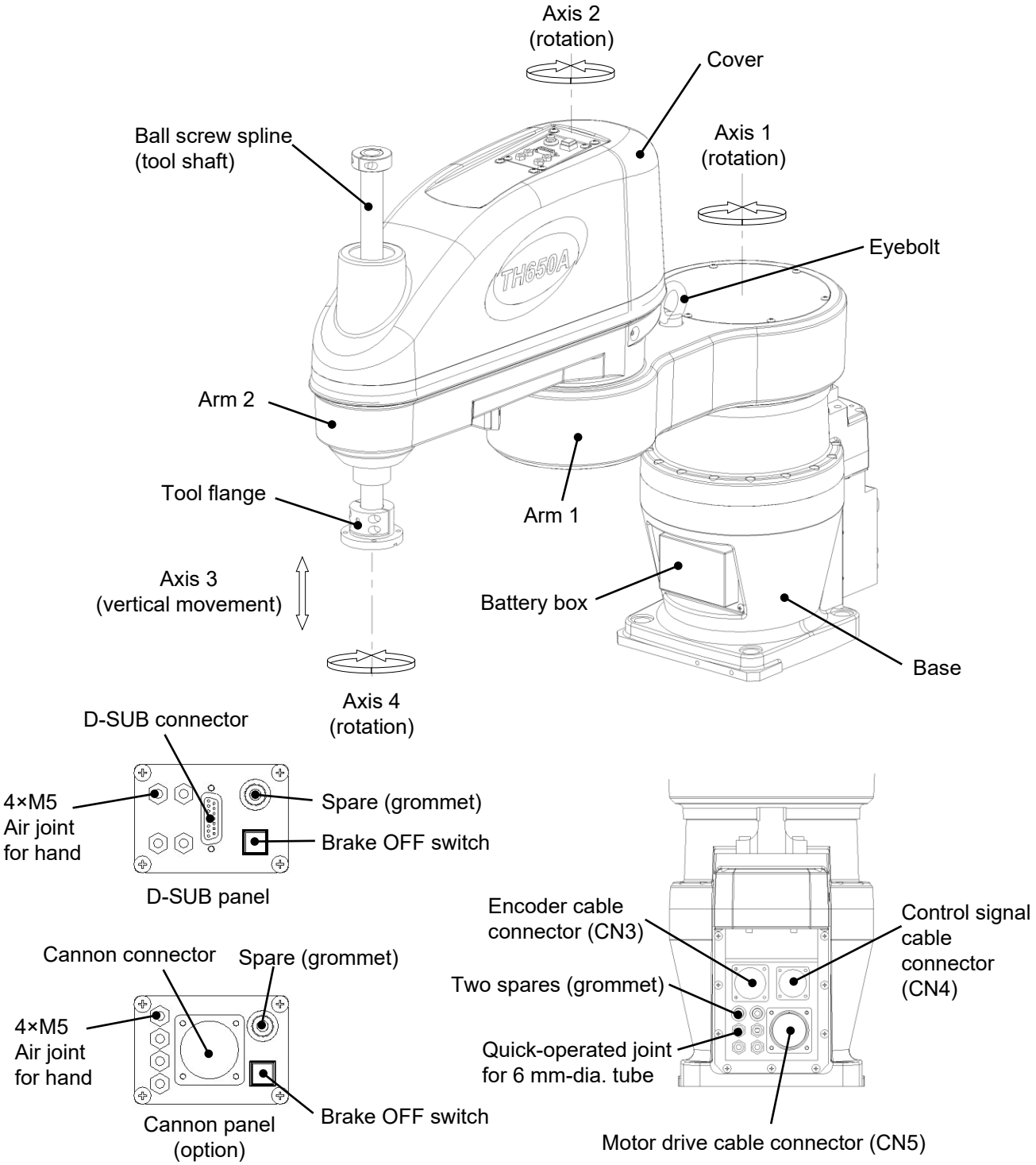


Fig. 1.1 Name of each part

1.2 External Dimensions

Fig. 1.2 refers to the external dimensions of the robot.

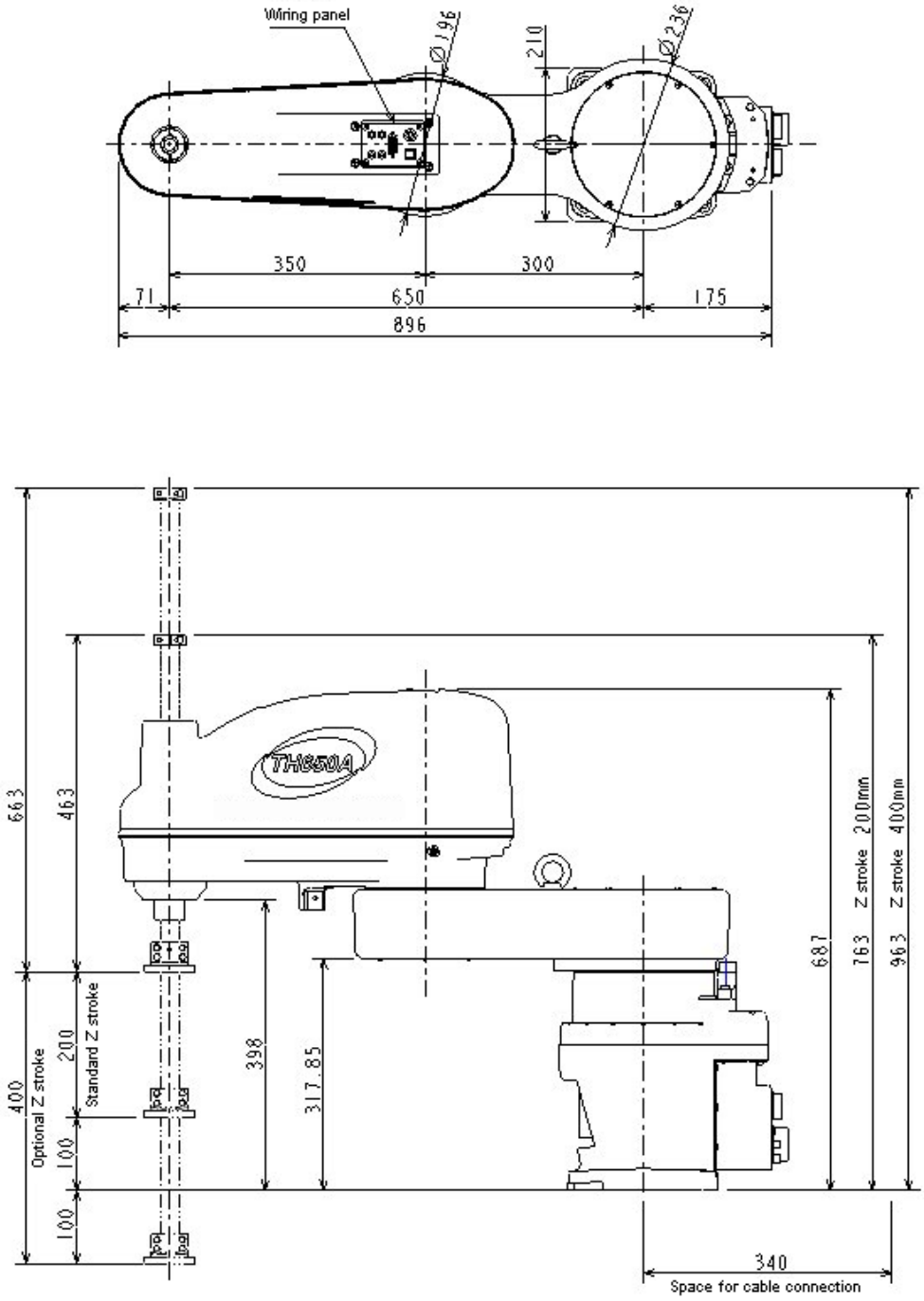


Fig. 1.2 External dimensions of the robot

1.3 Specifications Table

Item		Specifications
Structure		Horizontal multi-joint type SCARA robot
Model		TH650A
Applicable controller		TS2000
Mass of actuator		52 kg
No. of controlled axes		Four (4)
Arm length		650 mm (300 mm + 350 mm)
Motor capacity	Axis 1	1,000 (W)
	Axis 2	600 (W)
	Axis 3	400 (W)
	Axis 4	400 (W)
Operating range	Axis 1	±160 (deg)
	Axis 2	±143 (deg)
	Axis 3	200 (mm) [Option: 400 (mm)]
	Axis 4	±360 (deg)
Maximum speed (*1)	Axis 1	340 (deg/s)
	Axis 2	600 (deg/s)
	Axis 3	2,050 (mm/s)
	Axis 4	1,700 (deg/s)
	Composite speed of axes 1 and 2	7.52 (m/s)
Rated payload mass		2 (kg)
Maximum payload mass		10 (kg)
Permissible load inertia (*1)		0.1 ($kg \cdot m^2$)
Repeatability	X, Y	±0.01 (mm)
	Z	±0.01 (mm)
	C	±0.004 (deg)
Cycle time (*2) (When payload mass is 2 kg)		0.36 (sec)
Drive system		By means of AC servo motors
Position detection method		Absolute

*1: When the mass of load exceeds 2 kg, or when the gravity center position of load is away from the axis 4 center position, both the speed and acceleration should be reduced, using the PAYLOAD command.

*2: Shuttle time for rough positioning in horizontal direction of 300 mm and vertical direction of 25 mm.
Continuous operation in excess of effective load factor is not possible.

! CAUTION

Put the Z-axis (axis 3) in the raised position as much as possible, when moving Axes 1, 2, and 4. Moving Axis 1, 2, or 4 when the Z-axis is in low positions can lead to premature damage to the ball screw spline (Z-axis shaft).
If Axis 1, 2, or 4 must be moved while the Z-axis is in low positions due to unavoidable circumstances, prevent the ball screw spline vibrations by using the SPEED, ACCEL/DECEL, and PAYLOAD commands to adjust the operation speed and acceleration.
If Axis 1, 2, or 4 is moved while the Z-axis is in low positions, be extremely careful not to collide with any objects. Even if Axis 1, 2, or 4 is moved at low speed, a collision or other impact with an object can damage the ball screw spline (Z-axis shaft) before alarm occurs.

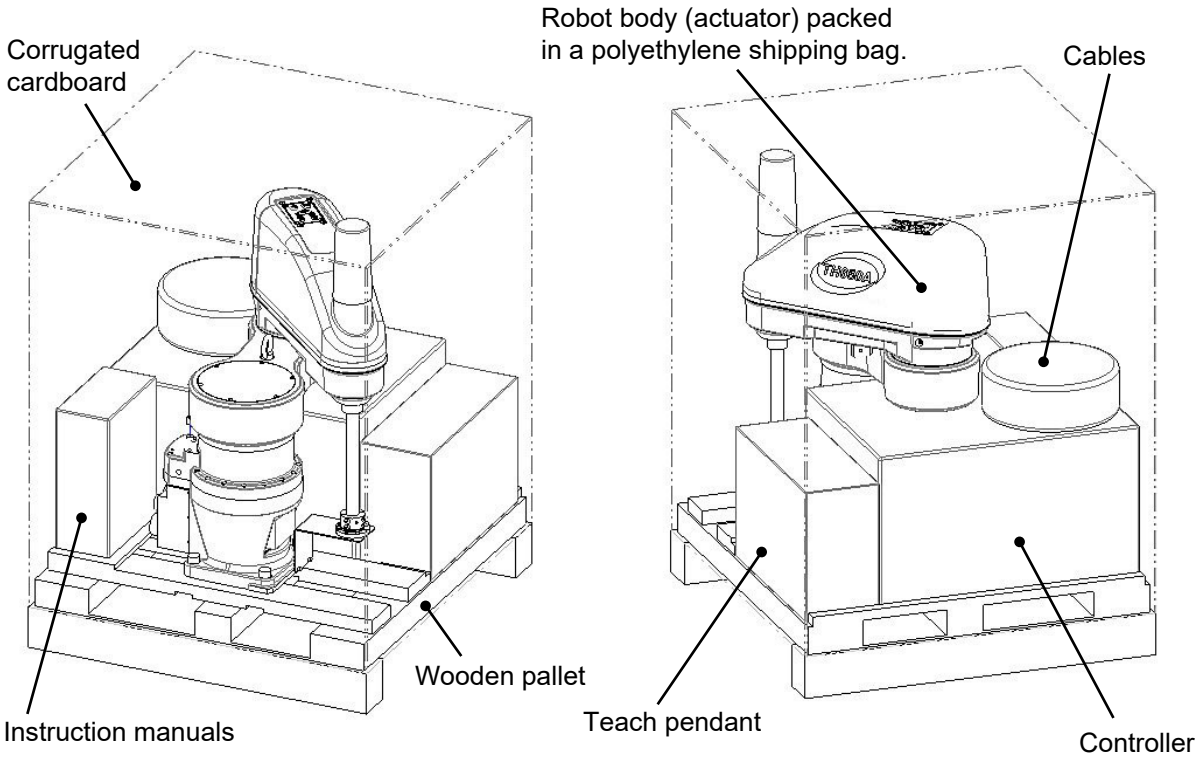
2. Transportation

2.1 Unpacking

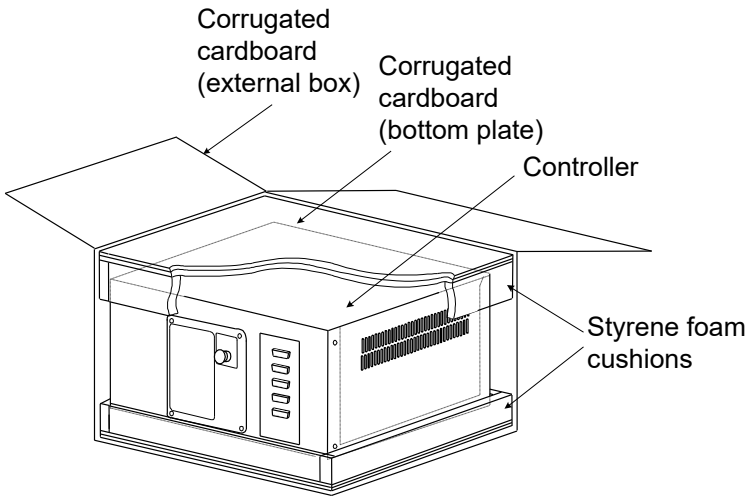
The robot and controller are shipped separately in corrugated cardboards. Fig. 2.1 shows each packaging state.

Open the packages in a location easily accessible, where the equipment is to be installed. Take careful precautions not to damage the robot and controller.

After opening the packages, make sure that all the accessories are present and that nothing has been damaged during transport.



Accessories included in the instruction manuals
System floppy disk: 3 pcs.
CD-ROM for TCPRGOS (option): 1 pc.
CD-ROM for TSPC (option): 1 pc.



Accessories included in the controller package
CN5 dummy connector: 1 pc.
CN6 dummy connector: 1 pc.
TP dummy connector: 1 pc.

Fig. 2.1 Packaging state

! DANGER

- If any parts of the equipment are found damaged or any accessories are missing after the shipment containing the robot and controller have reached your office, DO NOT install and operate them. Otherwise, the equipment will malfunction. Contact SHIBAURA MACHINE immediately.
- Dispose of the wooden pallet, corrugated cardboards, polyethylene shipping bags and cushion material according to the customer's in-house regulations.

2.2 Transportation

Move the robot and controller very carefully. Make sure that no excessive impact or vibration is exerted on the equipment. If the equipment is to be subject to vibration over a long period, be sure to tighten all the clamp and base set bolts completely.
If the equipment is to be moved to a location some distance from where it was unpacked, reposition the cushions as they were and put the equipment back into the corrugated cardboards.

2.2.1 Mass and Dimensions

The mass and outer dimensions of the robot are shown in Fig. 2.2.
For the mass and outer dimensions of the controller, see Fig. 3.5 of Para. 3.3.1.

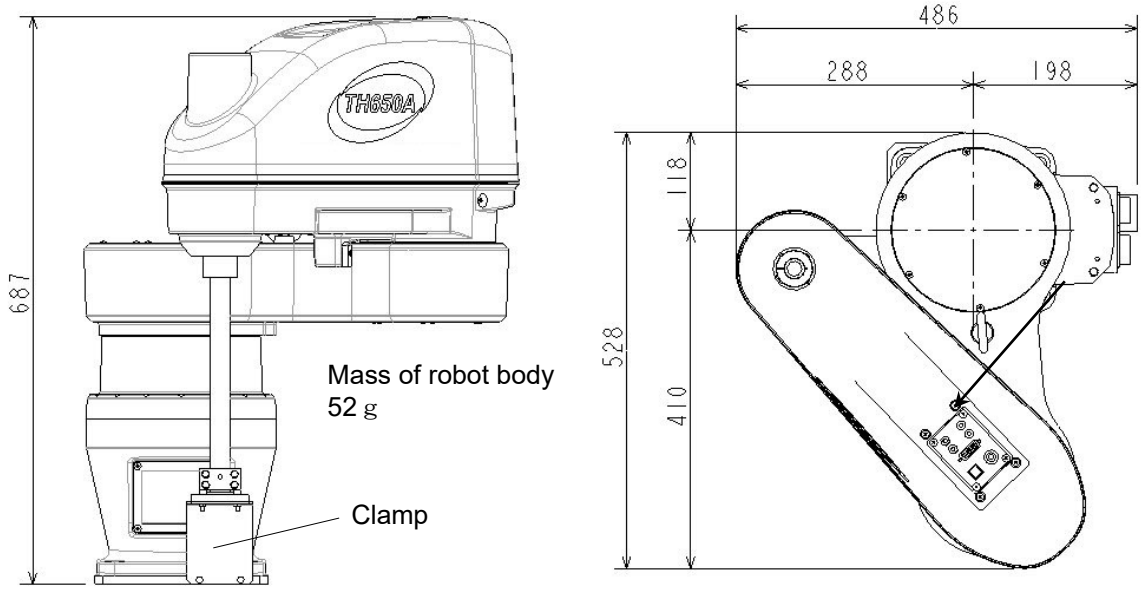


Fig. 2.2 Outer dimensions at transport

2.2.2 Transporting the Robot

In principle, the robot should be transported in the state shown in Fig. 2.2 above. Fold back and secure the arm with the attached clamp. (The robot is shipped in this posture. After you have unpacked the shipment, you should move it as it is.) At this time, take careful precautions not to impose a large force on the tool shaft.

! DANGER

- Be sure to secure the arm with the attached clamp before transporting the robot. Failure to do so could cause a hazardous situation as the arm will move when the robot is lifted.

It is possible to lift up and transport the robot. Pass the wire through the attached eyebolt, then lift up the robot carefully, as shown in Fig. 2.3.

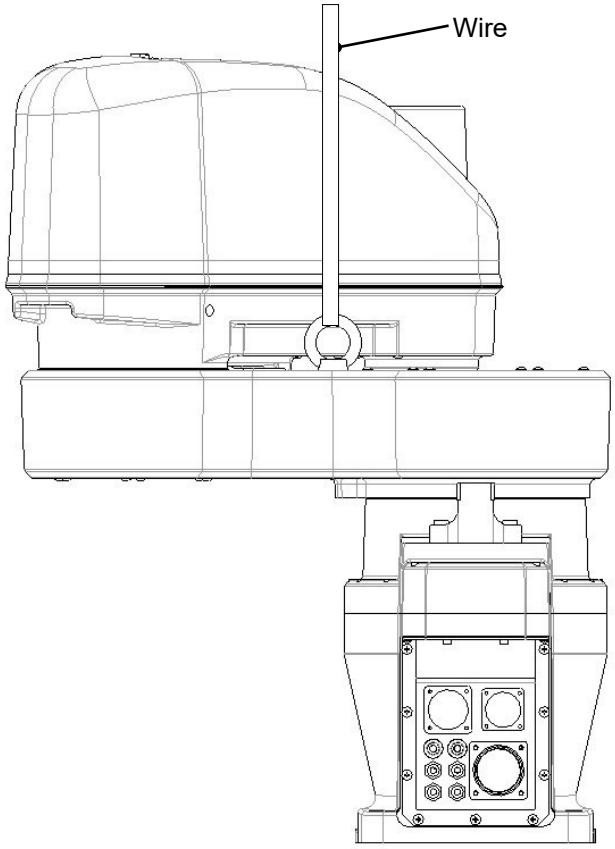


Fig. 2.3 Lifting up the robot

CAUTION

- The wire to be used should be such that can well withstand the mass of the robot.
- When lifting up the robot, it may tilt a little. Lift it up slowly.
- Lifting up and down should be performed carefully so that any impact cannot be exerted on the robot.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.

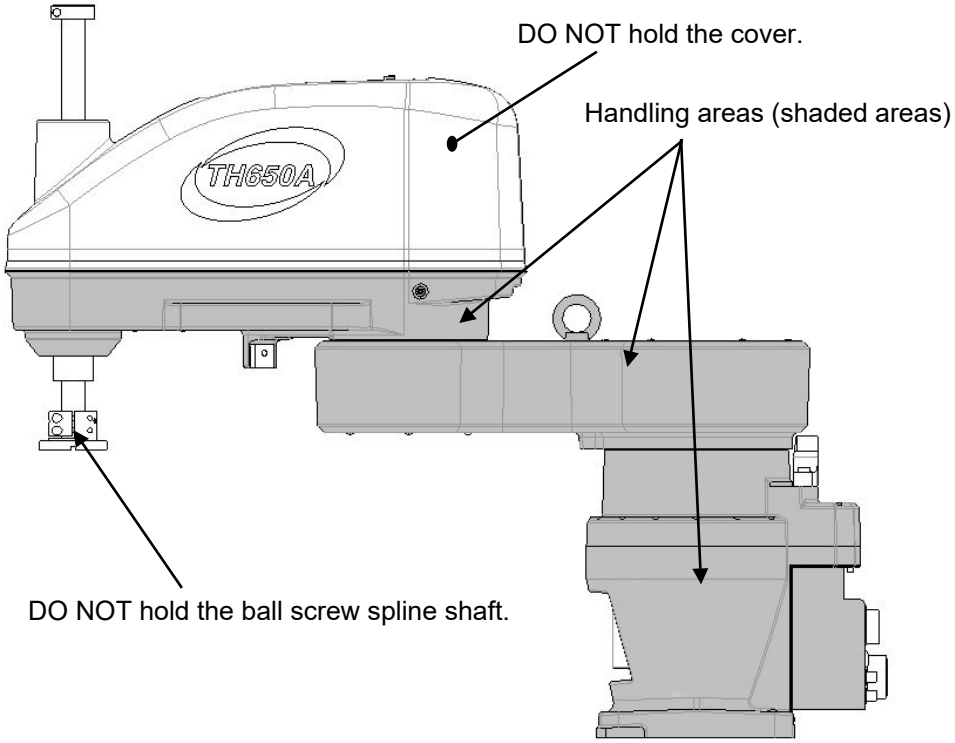


Fig. 2.4 Robot handling areas (shaded areas)

After the installation, remove the clamp and eyebolt used for transport.

! CAUTION

- When lifting up the robot by workers, hold the locations (shaded areas) by hands as shown in Fig. 2.4. If the arm 2, cover or ball screw spline shaft is held by hands, an unusually large force is exerted, resulting in a malfunction.
- When carrying the robot by workers, take careful precautions to prevent their hand or leg from being caught in the robot.
- The work should be performed by two (2) or more workers.

2.2.3 Transporting the Controller

Disconnect all cables and teach pendant before transporting the controller.

! DANGER

- When placing the controller on the floor, etc., make sure not to have your hands or feet caught.

2.3 Storage

Avoid storing the robot and controller for long periods of time after unpacking them. If this is unavoidable, however, strictly observe the following precautions for storage.

2.3.1 Storage Precautions for the Robot

! CAUTION

- Secure the base completely to prevent the robot from falling over. When placed directly on the floor, the robot is unstable and will fall over.
- Keep the robot out of direct sunlight. The timing belts and resin covers may deteriorate.
- Seal the robot in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture. As the tool shaft is susceptible to rust development, coat it with rust-preventive agent or grease the entire tool shaft beforehand.
- Before the use, apply the grease to the tool shaft.
- Before starting an operation, perform running completely.
- During storage, the life of the backup batteries will shorten. It is recommended to replace the batteries at the time of operation.

2.3.2 Storage Precautions for the Controller

! CAUTION

- Keep the controller out of direct sunlight. Otherwise, the controller interior will be excessively heated up, causing a trouble.
- Seal the controller in a vinyl bag to prevent rust development and contaminant. Put a desiccant in the bag to absorb moisture.

3. Installation

3.1 Installation Environment

Table 3.1 shows the environmental conditions for the location in which the robot and controller are to be installed.

Table 3.1 Environmental conditions for robot and controller

Item	Specifications
Temperature	In operation : 0 to 40°C In storage : –10 to 50°C
Humidity	20 to 90 % (Non-condensing) DO NOT install the robot where it may be subject to fluids such as water.
Altitude	1000 m or less
Vibration	In operation : 0.98 m/s^2 or less
Dust	No inductive dust should exist. Consult with SHIBAURA MACHINE first if you wish to use the robot and controller in a dusty environment.
Gas	No corrosive or combustible gas should exist.
Sunlight	The robot and controller should not be exposed to direct sunlight.
Power noise	A heavy noise source should not exist nearby.
Magnetic field	A heavy magnetic field source should not exist nearby.



DANGER

- Do not place the robot or controller near combustible. Doing so could lead to fires if it ignites due to a fault, etc.

⚠ CAUTION

- In the case where batteries for detecting the motor position are of alkaline type, the batteries can overheat, leak battery fluid, or rupture when used under high temperatures. Also, high temperatures can reduce the performance and lifespan of the battery. If using the robot under high temperatures, please consult with the SHIBAURA MACHINE sales office.
- When starting fast-movement operation in low-temperature environment, errors may occur because of the increased torque. In operating the robot in low-temperature environment, start by making low-speed continuous operation for a several minutes to soften the grease before starting high-speed operations.

3.2 Robot Installation

Before installing the robot, you should plan a layout, fully considering the working envelope (or operating range), coordinate system and space for maintenance.

3.2.1 External Dimensions

External view drawing of the robot is shown in Fig. 3.1.

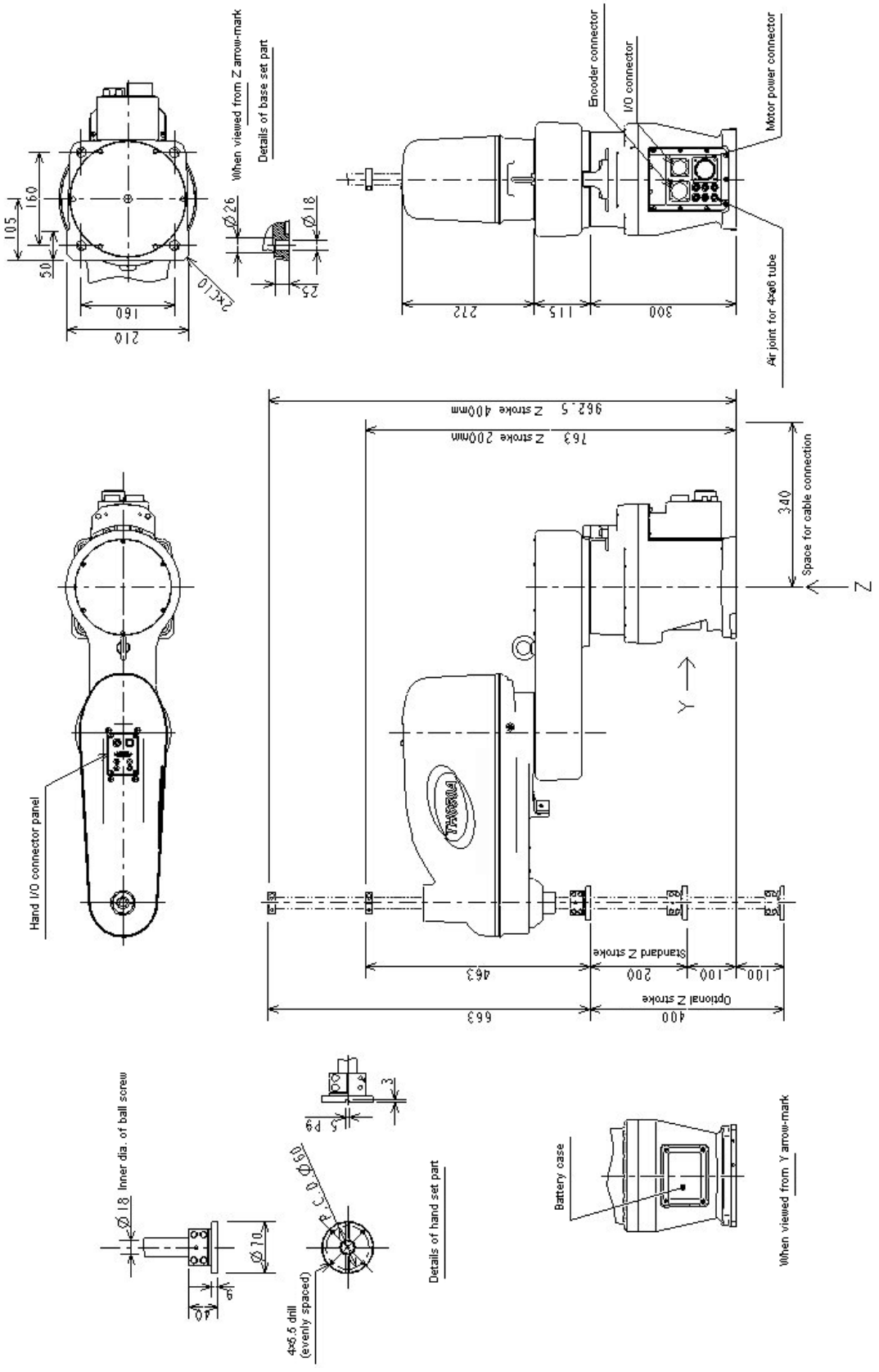


Fig. 3.1 External view

3.2.2 Working Envelope

Fig. 3.2 shows the working envelope of the robot.
Each axis can operate within the working envelope. To prevent the robot from moving out of the working envelope by mis-operation, the robot is equipped with mechanical stoppers outside the working envelope. Additionally, soft limits which can be set by the user are provided. For further information, see the user parameter instruction manual provided separately.

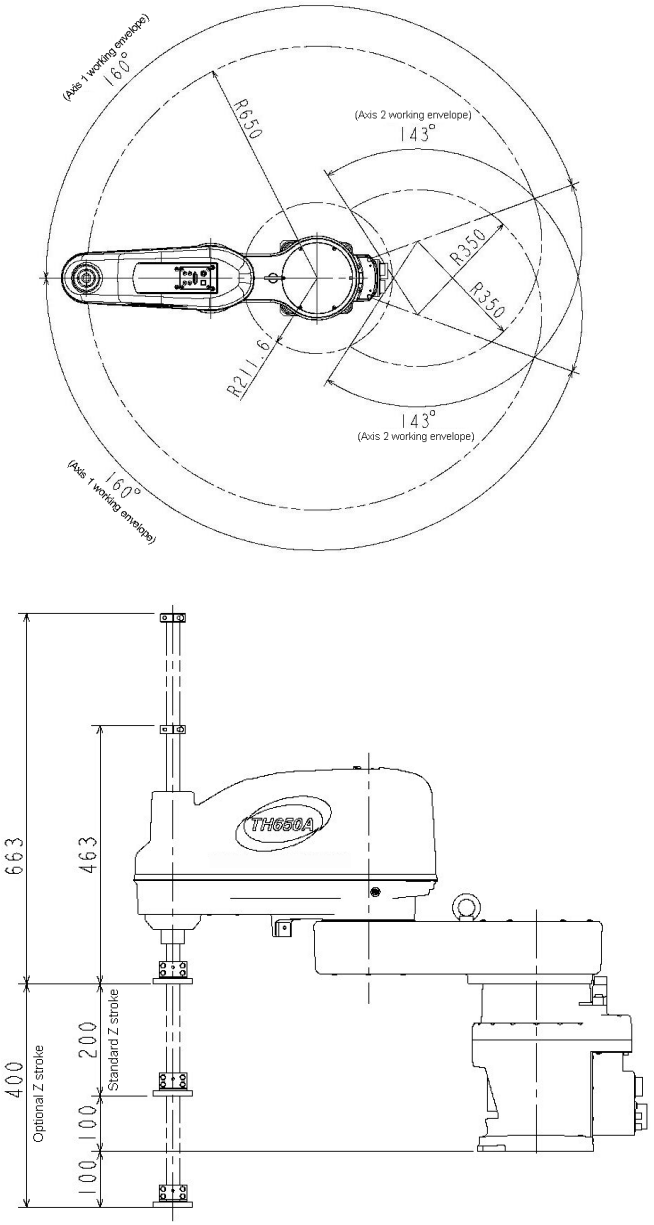


Fig. 3.2 Working envelope

3.2.3 Coordinate System

The robot's joint angle origin (0° or 0 mm position) is factory-calibrated according to the base reference planes. Fig. 3.3 shows the base coordinate system and origin of each axis joint angle.

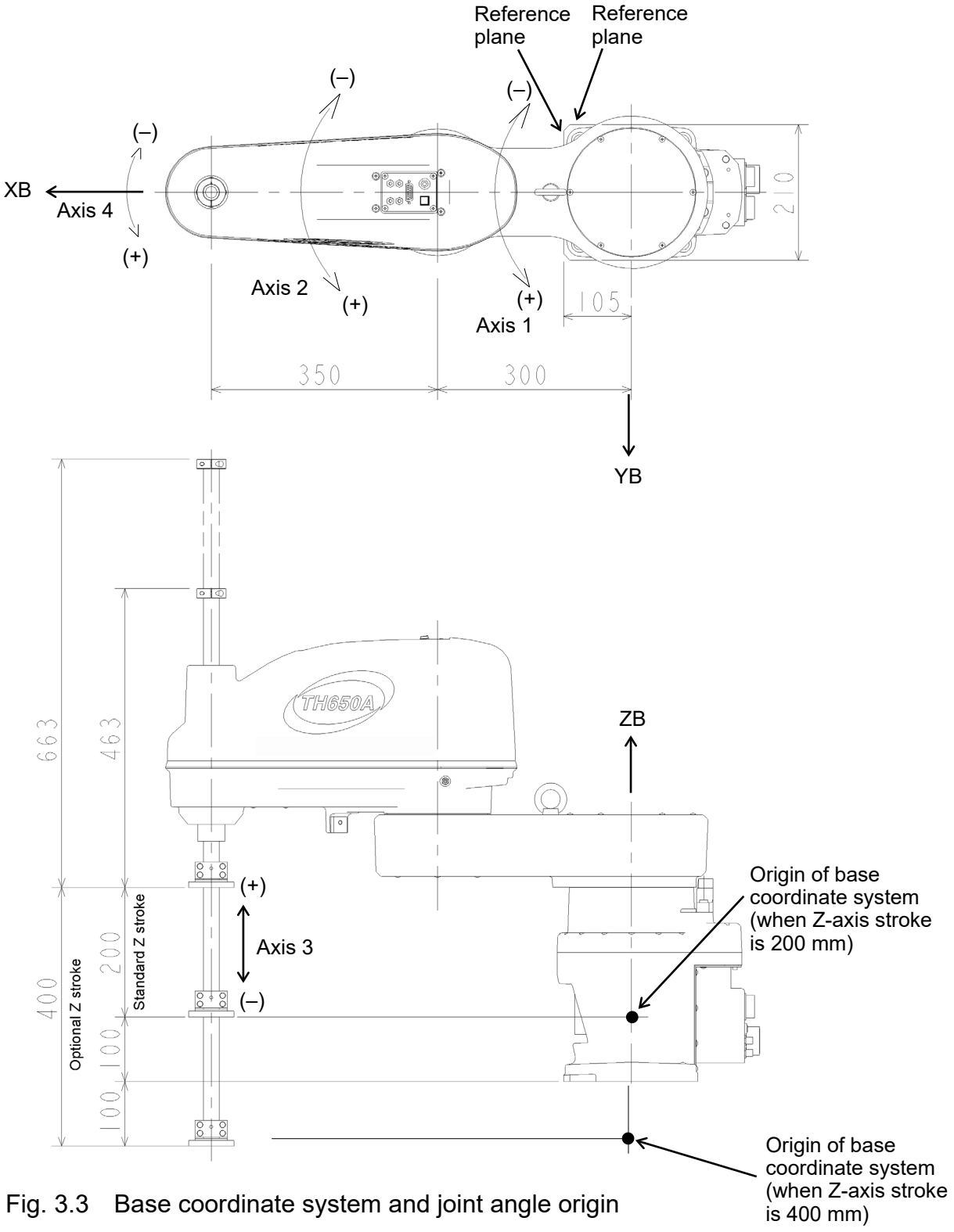


Fig. 3.3 Base coordinate system and joint angle origin

3.2.4 Installing the Robot

The robot is secured, using the set holes on the base (four (4) places). Use M16 hexagon socket head cap screws.
The robot installation method is shown in Fig. 3.4. Reference planes are provided on the base unit.
To align the robot position in the base coordinate system, or to replace the robot, provide adequate reference planes. Then, contact such reference planes to the base reference planes and secure the robot.

! CAUTION

- The robot will suddenly accelerate and decelerate during operation. When installing it on a frame, make sure that the frame has sufficient strength and rigidity.

  If the robot is installed on a frame that does not have sufficient rigidity, vibration will occur while the robot is operating, and could lead to faults.

  When installing the robot on the floor, secure the robot with anchor bolts, etc.
- Install the robot on a level place. Failure to do so could lead to a drop in performance or faults.

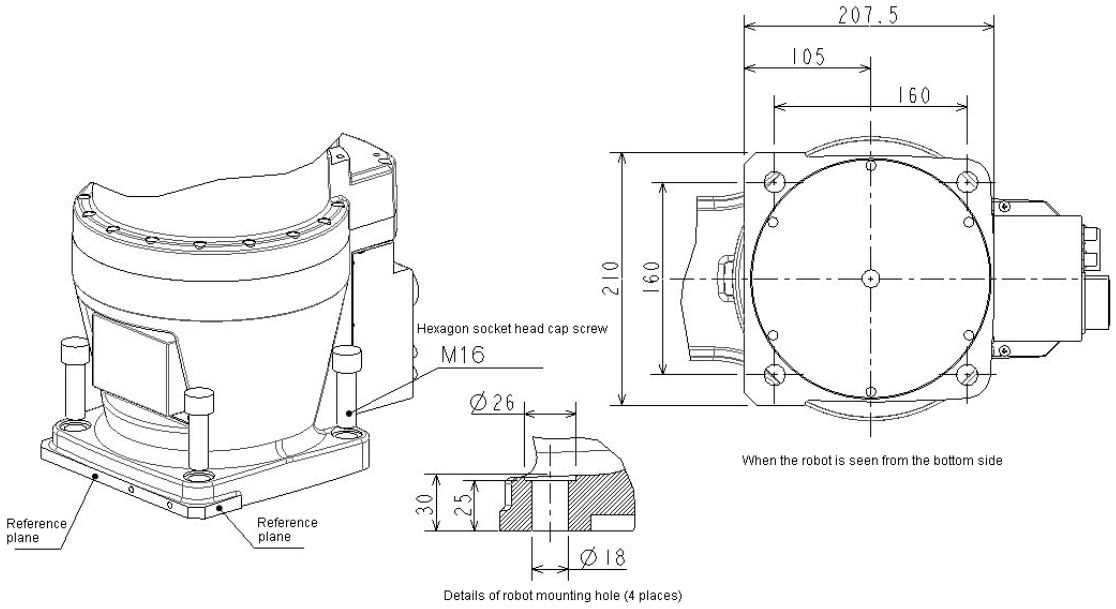
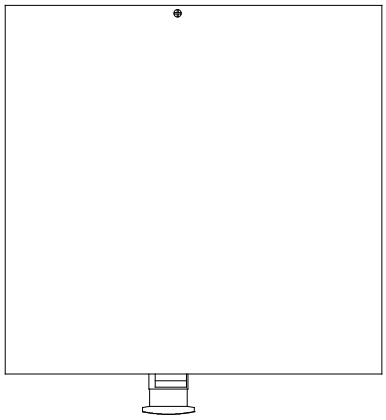


Fig. 3.4 Setting method

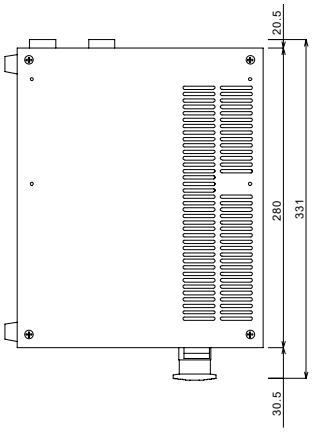
3.3 Installing the Controller

3.3.1 External Dimensions

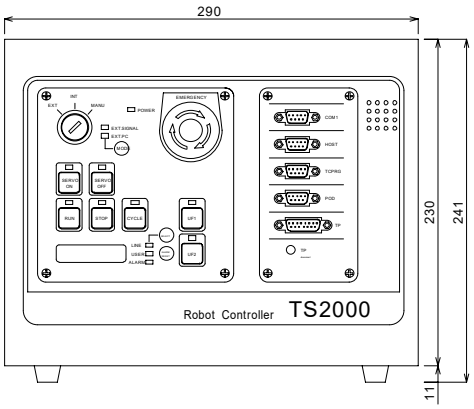
External view of the controller is shown in Fig. 3.5.



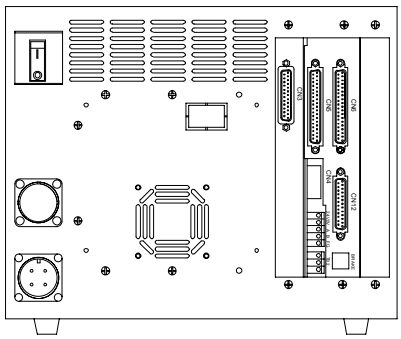
Top view



Left side view



Front view



Mass of controller:
12 kg

Fig. 3.5 External view of controller

3.3.2 Precautions for Direct Installation

It is necessary to provide a clearance of 50 mm or more in the horizontal direction and a clearance of 100 mm or more in the upward direction near the controller.
When installing the controller lengthwise, be sure to locate the right lateral side for sideways installation at the bottom side. (The lengthwise installation is an option.)

! CAUTION

- Provide a ventilation space at the side of the controller so that the air vent holes are not blocked. The space equal to the length of the legs should be kept below the lower surface.

  Failure to do so could cause the cooling performance to drop or to faults.
- DO NOT stack the controllers.
- DO NOT place any object on top of the controller.

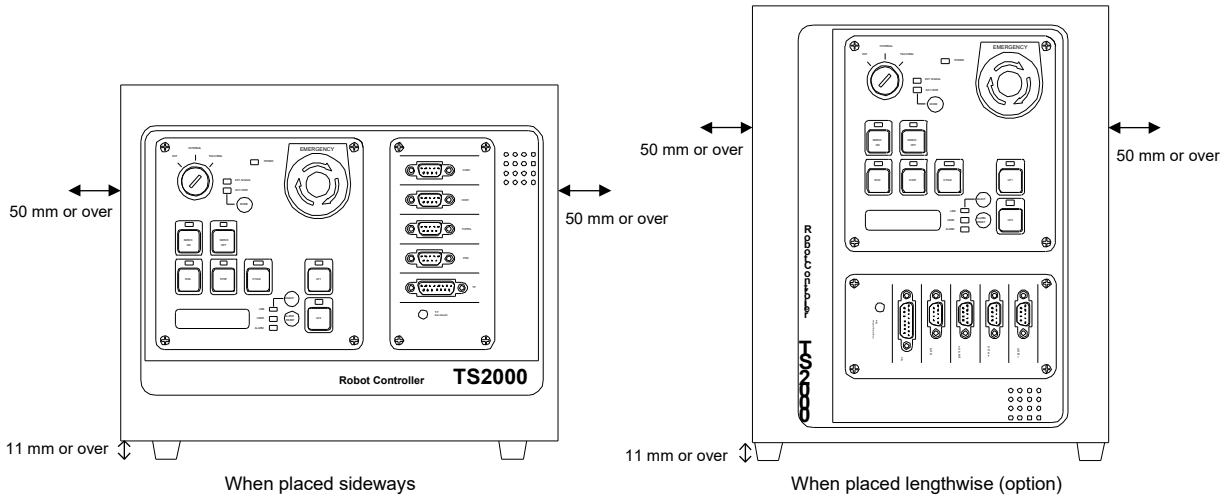


Fig. 3.6 Controller ventilation space

3.3.3 Rack Mounting Dimensions

When mounting the robot controller in a rack, set the side brackets using the screw holes provided on both ends of the front panel, and secure the controller to the rack. The side brackets [1] in Fig. 3.7 are optional.

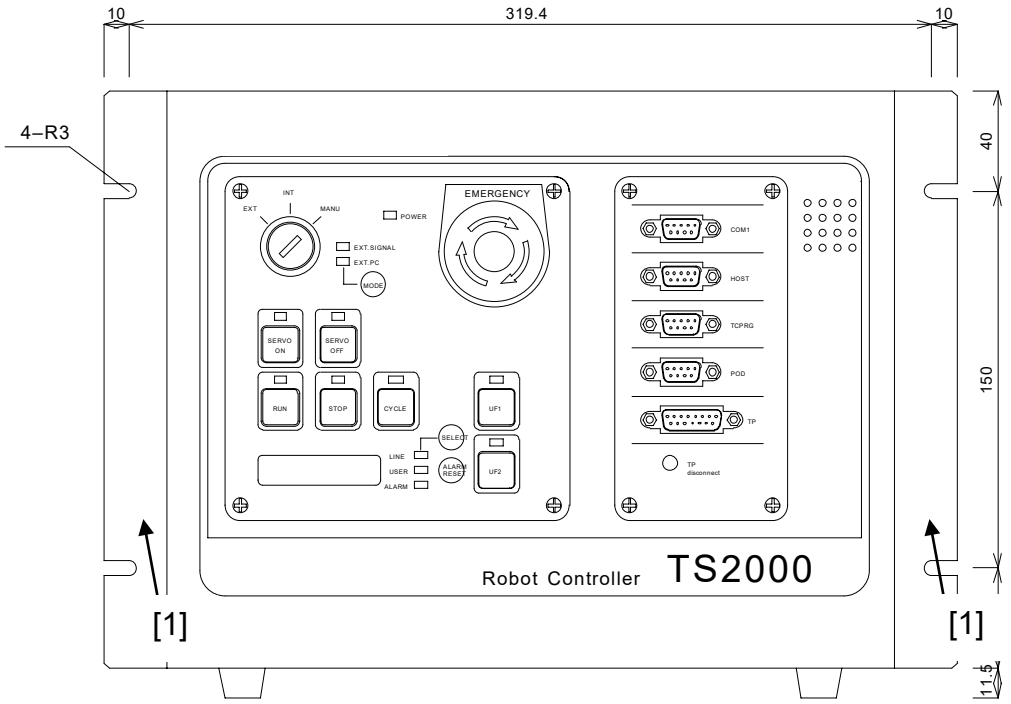


Fig. 3.7 Screw hole dimensions for securing controller

3.3.4 Precautions for Rack Mounting

Pay attention to the following matters when rack-mounting the robot controller.

a) When mounting the robot controller in a rack, use the screw holes provided on both ends of the front panel, and secure the controller. (Optional side brackets are required.)

! CAUTION

- If the rack is completely sealed, provide holes to allow the air to be let out, force-ventilate the rack with a fan, or cool it indirectly so that the heat will not be trapped in the rack.

  If the heat is trapped in the rack or controller, faults could occur.

b) As the cable connectors are connected to the rear side of the controller, provide a space of 110 mm or over on the rear side.
For maintenance, the upper cover should be removed. (See Fig. 3.8.)

Keep this in mind when installing the controller. At maintenance, the controller should be removed from the rack. Specifically, be careful of the following points.

1) Arrange the cables around the rear side of the controller (so that the controller can be removed).
2) Arrange the cables between the controller and control panel when the control panel is separated.
3) Connect all cables in such a manner that the robot can be operated even if the controller is removed from the rack.

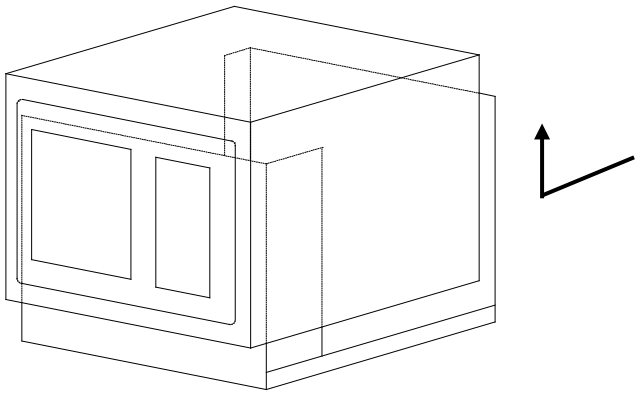


Fig. 3.8 Removing upper cover

c) When the controller is mounted on the rack, the weight of the controller should be supported with the legs of the controller. Screw holes for rack-mounting the controller are used for securing the controller panel, and the weight of the controller cannot be supported only with these screws.

d) On the front of the controller, a clearance of approx. 90 mm should be provided for connecting the connector of the teach pendant. Even if the teach pendant is not used, a clearance of approx. 60 mm is required for connecting a dummy plug.

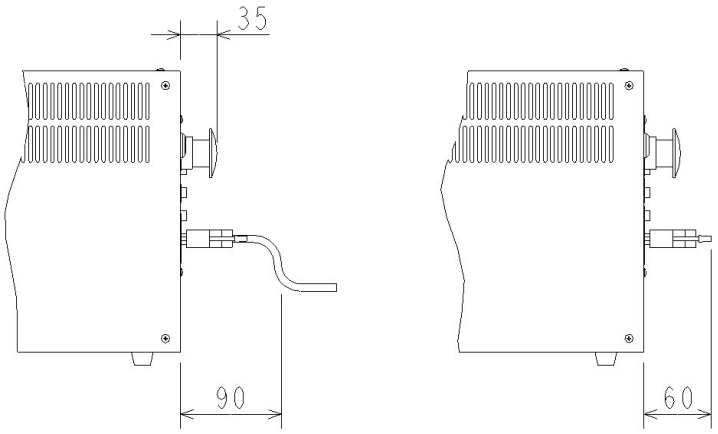


Fig. 3.9 Clearance of controller front side

3.4 Precautions for Handling the Teach Pendant

Be careful of the following matters when handling the teach pendant.

! CAUTION

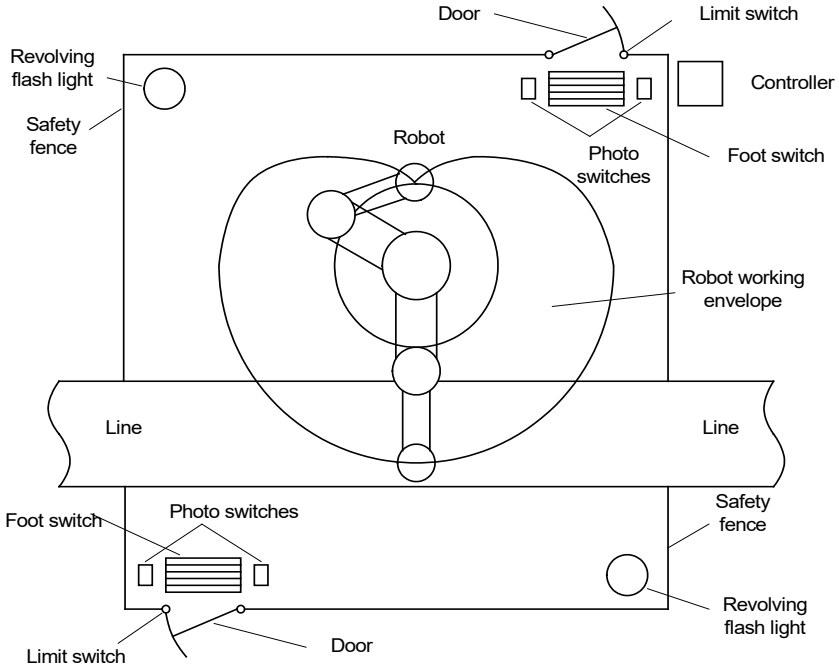
- DO NOT drop the teach pendant or hit it against anything.
- DO NOT pull the cable running from the teach pendant.
- DO NOT press the switches on the teach pendant with anything sharp (like the tip of a knife, pencil, ball-point pen, etc.).
- DO NOT place or use the teach pendant near open flames.
- DO NOT leave the teach pendant in direct sunlight for a long period of time.

3.5 Safety Measures

a) When installing the robot, provide sufficient space to carry out the work safely.

b) Clarify the hazard zone, and provide a safety fence so that other persons cannot enter the zone easily. The hazard zone is the zone near the robot's working space where a hazardous state could occur if a person enters.

c) Provide limit switches, photo switches, foot switch, etc., at the entrance of the safety fence to provide an emergency stop function that will stop the robot if a person enters the hazard zone. The emergency stop function should be an electrically independent normal close contact (closed in normal operation) with compulsive opening function and must not be automatically recovered.



d) The controller should be installed at a place outside the hazard zone where the operator can view the robot movement.

4. System Connections

4.1 Cable Wiring

This section describes the various types of cables and connectors and explains how these are to be connected.

4.1.1 Connector Arrangement on the Controller

The cables connected to the robot controller are shown in Fig. 4.1.

[1] Power cable
[2] Motor cable (CN2)
[3] Encoder cable (CN3)
[4] Robot control signal cable (HAND I/O) (CN4)
[5] External operation input signal cables (CN5, 6, 12)
[6] Brake signal cable
[7] Distribution I/O cable

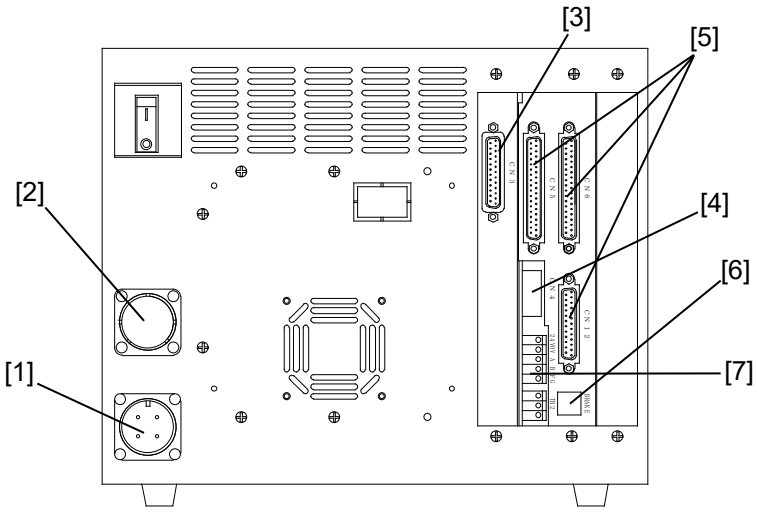


Fig. 4.1 Robot controller connector arrangement

In the subsequent paragraphs, we explain how to connect cables [1] to [4] inclusive. For information on how to connect cables [5], [6] and [7], refer to the Interface Manual.

4.1.2 Connecting the Power Cable “CN1”

([1] of Fig. 4.1; plug connector attached)

The power cable is used to supply the main AC power to the controller.

Table 4.1 Power supply specifications

Item	Specifications
Power supply	Single phase, AC 190 ~ 250 V, 50/60 Hz ±1 Hz
Power capacity	4.4 kVA
Instantaneous power failure	Within 40 msec
Grounding	Grounding with grounding resistance of 100 Ω or less

The connector is CN1 ([1] of Fig. 4.1).

CN1 plug connector	Type: JL04V–6A18–10SE–EB	Maker: Japan Aviation Electronics Industry
CN1 cable clamp	Type: JL04–18CK	Maker: Japan Aviation Electronics Industry
Wire	3.5 mm^2 ~ 5.5 mm^2	

As the cable is not an accessory, use the attached plug connector connected to CN1 on the controller side to manufacture a cable.
Wires are to be soldered to the connector.

! DANGER

- Be sure to use the designated wire. Failure to do so could lead to fires or faults.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

For the terminal arrangement, see Para. 4.1.7.

! CAUTION

- Unless the main power is normally supplied to the controller due to phase defect or voltage drop, an error of “8–027 Slow Charge error” occurs. When this happens, make sure that the maser power voltage at the controller power connector satisfies the specified input power of the controller, and that the same voltage is stabilized.
- For details of the 8–027 error, see Para. 13.7 of the Operator’s Manual.

4.1.3 Connecting the Motor Cable “CN2”

([2] of Fig. 4.1) (Cable attached)

The motor cable connects the controller and robot, and supplies the power required to rotate the motor from the controller servo driver to each axis feed motor of the robot. The connector for connecting the motor cable is CN2 ([2] of Fig. 4.1). Location of the motor cable on the robot side is [1] in Fig. 4.2.

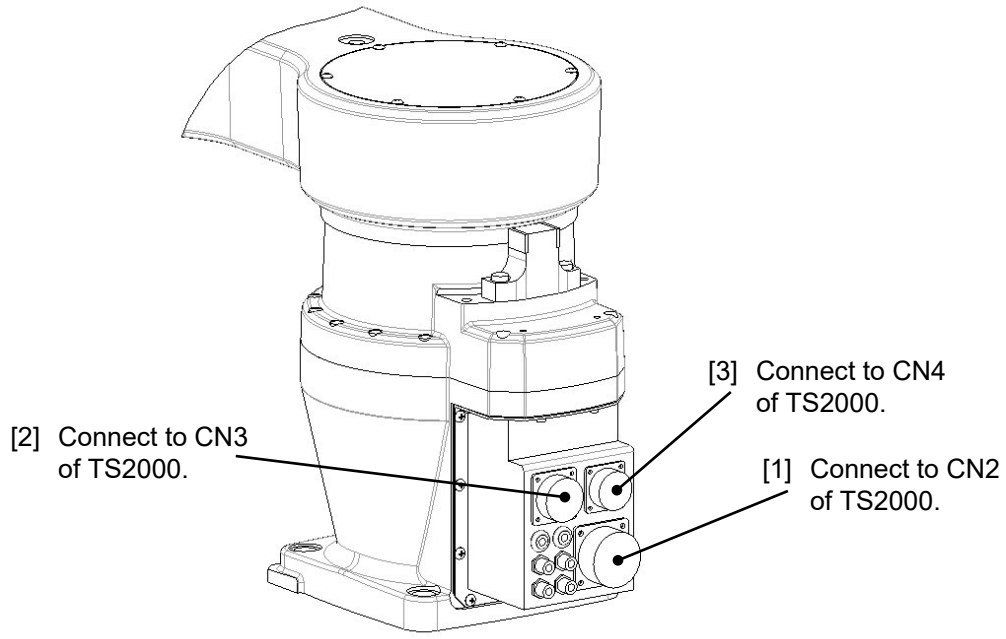


Fig. 4.2 Robot side connector arrangement

4.1.4 Connecting the Encoder Cable “CN3”

([3] of Fig. 4.1) (Cable attached)

The encoder cable is a signal line used to transmit a signal from the rotation angle detecting encoder of each robot axis to the controller.
The connector for connecting the encoder cable is CN3 ([3] of Fig. 4.1). Location of the encoder cable on the robot side is [2] in Fig. 4.2.

4.1.5 Connecting the Robot Control Signal Cable “CN4”

([4] of Fig. 4.1) (Cable: option)

The robot control signal cable is used for motor brake ON and OFF, and input and output of robot control signals such as hand operation signal.
The connector for connecting the robot control signal cable is CN4 ([4] of Fig. 4.1).
Location of the robot control signal cable on the robot side is [3] in Fig. 4.2.

4.1.6 Connecting and Disconnecting Cables

! CAUTION

- Before connecting or disconnecting any controller cable, be sure to turn off the main power (“POWER”) switch.
- When disconnecting a cable, be sure to pull the plug and not the cord. Otherwise, you may damage the cable.

a) Circular connectors: CN1, CN2
Firstly align the key position, and completely insert the connector on the cable side into the controller connector. Then turn the cable side lock screw to the right to clamp the cable. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screw is clamped completely.
To disconnect the connectors, turn the lock screw to the left and pull out the cable side connector.

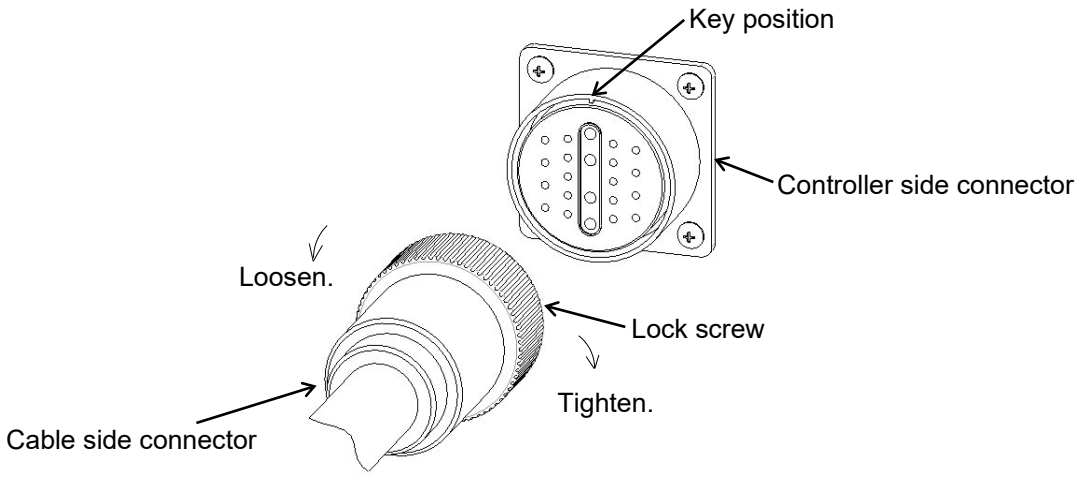


Fig. 4.3 Connecting and disconnecting a circular connector

b) Square connectors: CN3, CN4, CN5, CN6, CN12
Firstly, completely insert the cable side connector into the controller connector. Then tighten the lock screws on both ends of the cable side connector with a screwdriver. A loose screw can cause a contact failure or other accident. To avoid this, make sure that the screws are clamped completely
To disconnect the connectors, first loosen the lock screws, then pull out the cable side connector. CN4 is a quick-operated lock type connector, however.

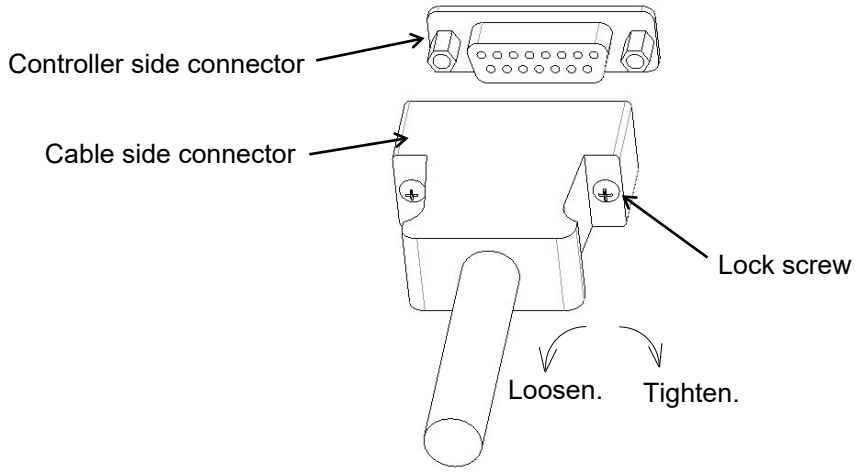
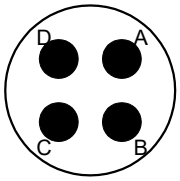


Fig. 4.4 Connecting and disconnecting a square connector

4.1.7 Examples of Connector Terminal Arrangement

a) Power cable connector CN1



Connects to controller.

Type: JL04V–2E18–10PE–B

Manufacturer:
Japan Aviation
Electronics Industry

A
B
C
D

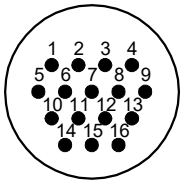
Single phase, AC 190~250 V, 50/60 Hz

Grounding with grounding resistance of 100 Ω or less.

! DANGER

- Completely connect the grounding cable. Otherwise, an electric shock or fire may be caused if a fault or electric leak occurred. Or mis-operation may be caused by noise.

b) Motor cable connector CN2

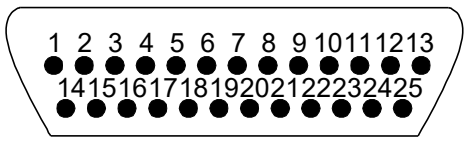


Connects to controller.

Type: JR25RK–16S

Manufacturer: Hirose Electric

c) Encoder cable connector CN3

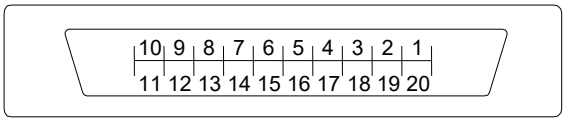


Connects to controller.

Type: XM2A–2501

Manufacturer: OMRON

d) Robot control signal cable connector CN4

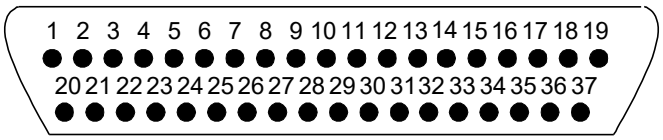


Connects to controller.

Type: PCR–E20LMD

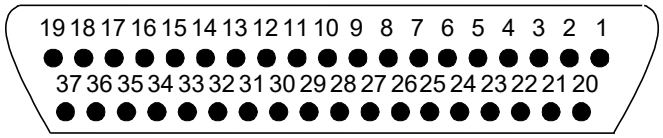
Manufacturer: Honda Tsushin Kogyo

e) External input signal cable connector CN5



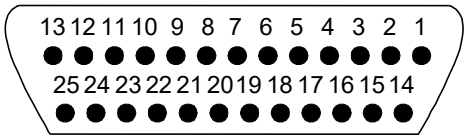
Connects to controller.
Type: XM2C–3712–112
Manufacturer: OMRON

f) External output signal cable connector CN6



Connects to controller.
Type: XM3B–3722–112
Manufacturer: OMRON

g) External input/output signal cable connector CN12



Connects to controller.
Type: XM3B–2522–112
Manufacturer: OMRON

4.2 Controller Connector Signals

4.2.1 Connector Signal Connection Diagrams

Diagrams showing which signals correspond to which terminals are shown in Section 2 of the Interface Manual.

4.2.2 Jumpers for Safety Related Signals

The following system input signals are provided to serve for the safety purpose.

System input signals	CN5–16	(STOP)
	CN6–34	(SVOFF)
	CN5–17	(BREAK)
	CN6–16, 35	(EMS2B ~ EMS2C)
	CN6–17, 36	(EMS1B ~ EMS1C)

These signals are already jumpered for the connectors provided for the controller. If you wish to use or change them, therefore, you should remove the jumpers and rewire as appropriate. If you plan to use the robot without using system input signals, be sure to connect the attached connectors to the controller side CN5, CN6 connectors. Unless the following signals are used as the system signals, jumper them also.

CN5–36 (LOW_SPD)
CN5–35 (CYCLE)

Connector jumpers

CN5		CN6	
16–18	17–18	18–34	–
(35–37)	(36–19)	16–35	17–36

! CAUTION

- Unless the signals of SVOFF and emergency stop contacts 1, 2 are jumpered, the controller servo power cannot be turned on.
- Unless the CYCLE signal is jumpered, the controller enters the cycle operation mode.
- Unless the LOW_SPD signal is jumpered, the robot is operated at low speed during automatic operation.
- Unless the STOP signal is jumpered, automatic operation of the robot is not possible.
- Unless the BREAK signal is jumpered, automatic operation of the robot is not possible.

4.3 Separating Control Panel from Controller

4.3.1 Removing Control Panel

Remove the control panel in the following manner.

a) Loosen the four (4) screws at the four (4) corners, which secure the control panel.

b) Remove these screws, then carefully draw out the control panel toward your side.

Caution: Be careful of the cable connected on the rear side.

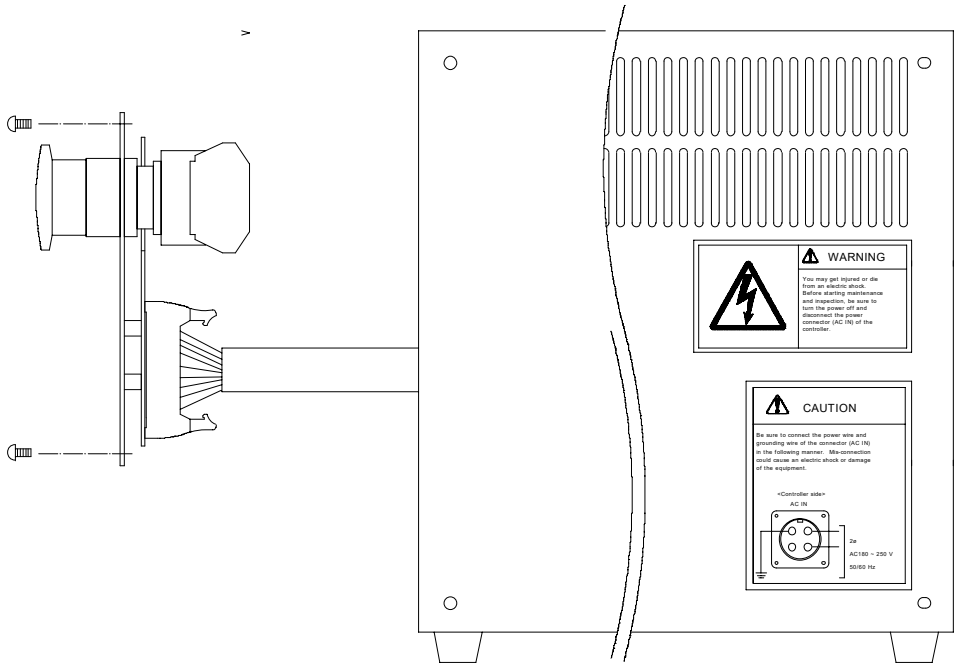


Fig. 4.5 Removing control panel

4.3.2 Cable between Controller and Control Panel

The cable required to connect the control panel and controller when they are installed separately can be provided optionally.

4.3.3 Control Panel Mounting Dimensions

The dimensions of mounting the control panel are shown in Fig. 4.6.
Cross truss head screws (ø3×6, ZN3–B) are used.

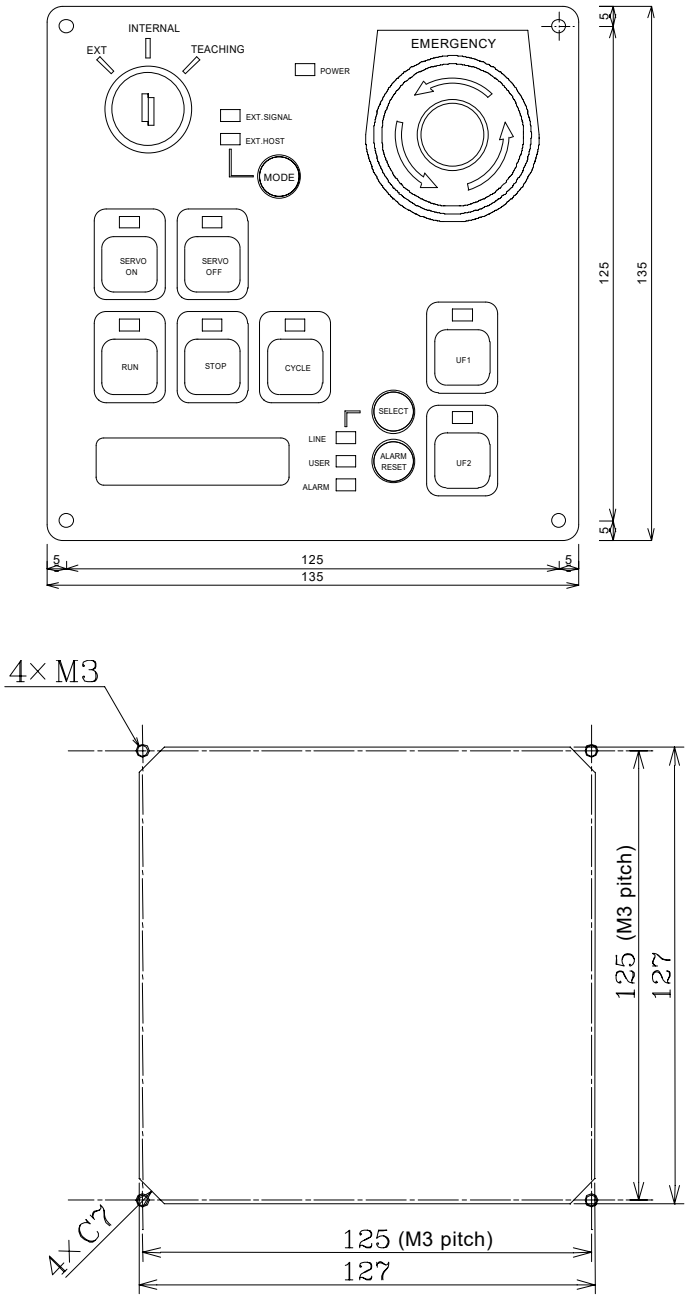


Fig. 4.6 Control panel mounting dimensions

4.3.4 Mounting Dummy Panel on Controller

When the control panel has been disengaged from the controller, mount a dummy panel on the place where the control panel was set before, as shown in Fig. 4.7. The dummy panel, set parts, etc. are provided optionally.

a) Connect the cable connector which was disconnected when separating the controller from the control panel, to the rear side of the dummy panel, then screw both ends of the connector. When mounting the connector, use the cross truss screws (ø3×6, ZN3–B).

b) Screw the dummy panel into the controller.

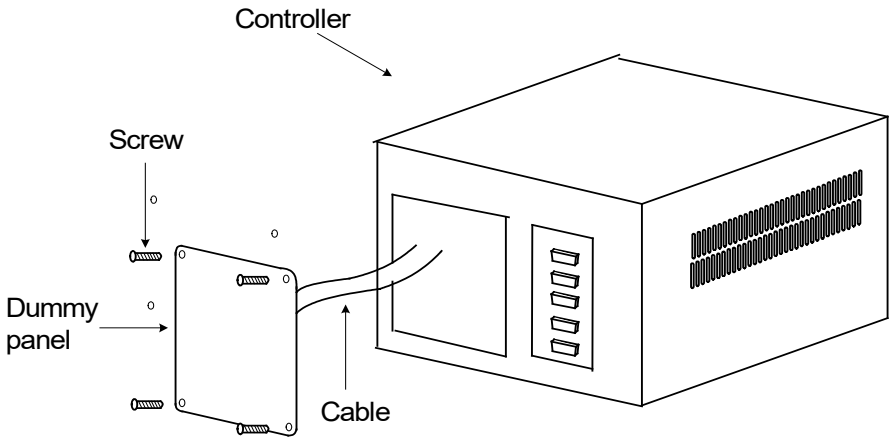


Fig. 4.7 Mounting dummy panel

4.3.5 Dimensions when Separating Control Panel

Fig. 4.8 shows the connections of the control panel and dummy panel. Provide a clearance of 50 mm or more (with cover, 60 mm or more) on the rear side of the separated control panel.
When the cable is connected to the dummy panel of the controller, provide a clearance of 80 mm or so in front of the controller as the cable connector sticks out of the panel surface.

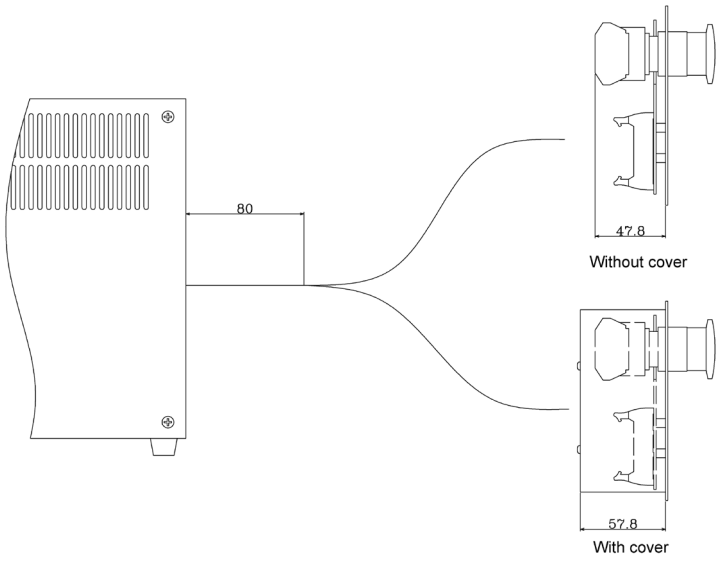


Fig. 4.8 Cable connections of dummy panel and control panel

5. Tool Interface

5.1 Mounting Tool

Mount the tool on the tool set flange at the end of the tool shaft. The dimensions of the tool set flange are shown in Fig. 5.1.
As shown in Fig. 5.1, the tool is centered with the ø20H7 mating section. The tool direction is adjusted by means of the 5 × 5 keys and secured with four (4) M5 bolts.

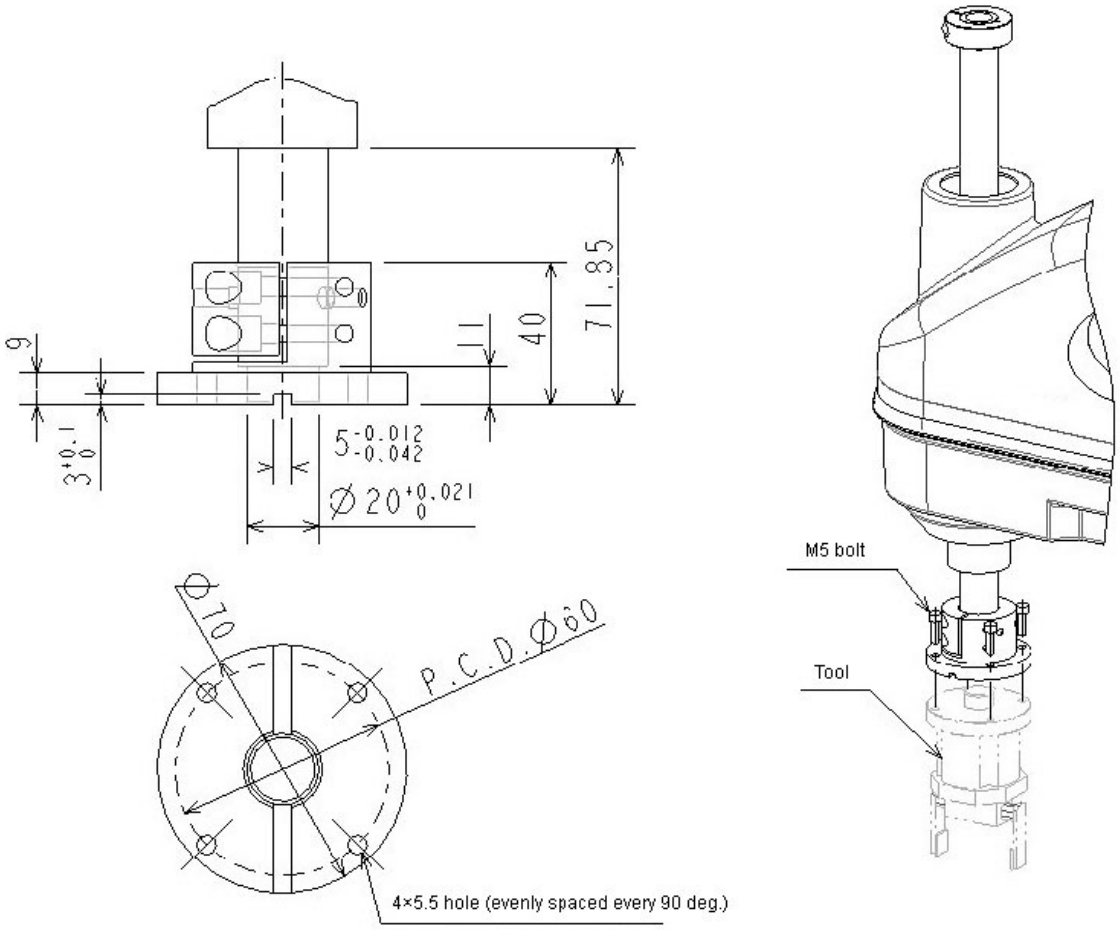


Fig. 5.1 Tool mounting dimensions

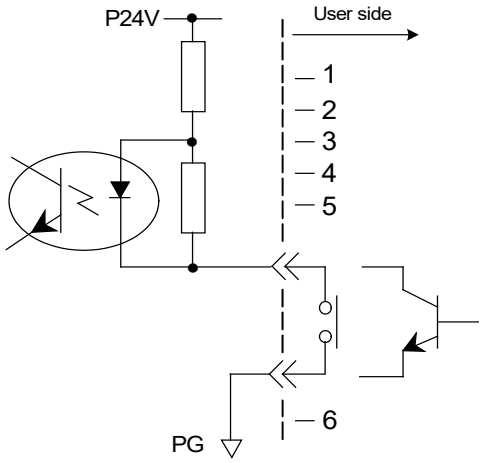
5.2 Tool Wiring and Piping

The robot is provided with wiring and air piping for the tool. These wiring and piping extend to the arm 2 and are used as follows:

5.2.1 Tool Signals (Controller Side)

The controller is provided with tool signals (i.e., five (5) input signals for sensor, etc., four (4) control signals for solenoid valve, etc., DC 24 V and DC 24 VGND signals), which can be connected also from the controller side. (A cable between the controller and cable is an option.) The relevant signals are described below.

a) Input signal connector CN4

Pin	Signal name		Signal No.	Input circuit and example of connections
5	GRP1 OPN	Grip 1 open	201	
6	GRP1 CLS	Grip 1 close	202	
7	GRP2 OPN	Grip 2 open	203	
8	GRP2 CLS	Grip 2 close	204	
9	WORK	Workpiece	205	
20	PG	0 V		
Case	Shield			

As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: DC24 V, 10 mA or over (circuit current: approx. 7 mA)

Minimum contact current: DC24 V, 1 mA

Contact impedance: 100 Ω or less

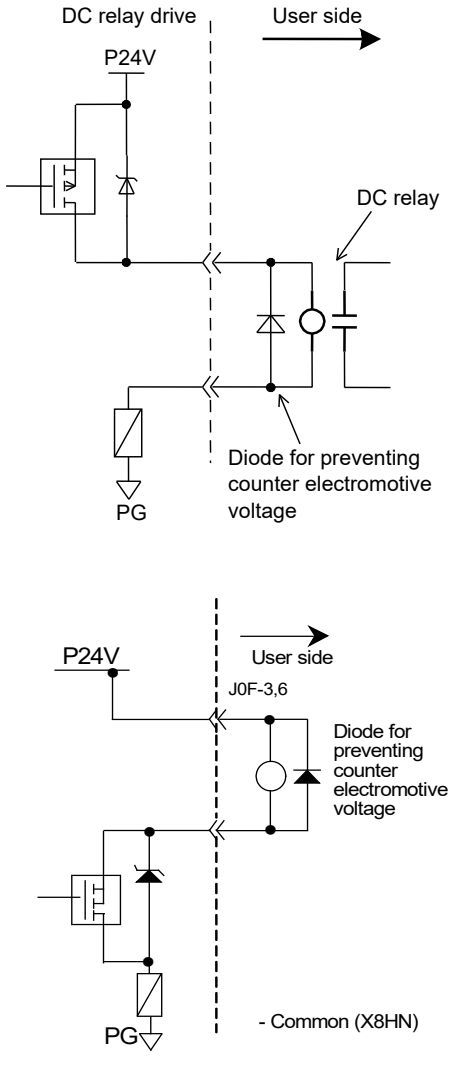
Transistor specifications:

Withhold voltage between collector and emitter: 30 V or over

Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)

Leak current between collector and emitter: 100 μA or less

b) Output signal connector CN4

Pin	Signal name		Signal No.	Output circuit and example of connections
11	GRP1	Grip 1	201	
12	$\overline{\text{GRP1}}$	– Grip 1	202	
13	GRP2	Grip 2	203	
14	$\overline{\text{GRP2}}$	– Grip 2	204	
17	P24V		DC24 V power	
18	P24V		DC24 V power	
19	P24V		DC24 V power	

By using the DC24 V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (PG) of the robot controller. NEVER connect +24 V with P24 V. Otherwise, the power supply may be damaged.

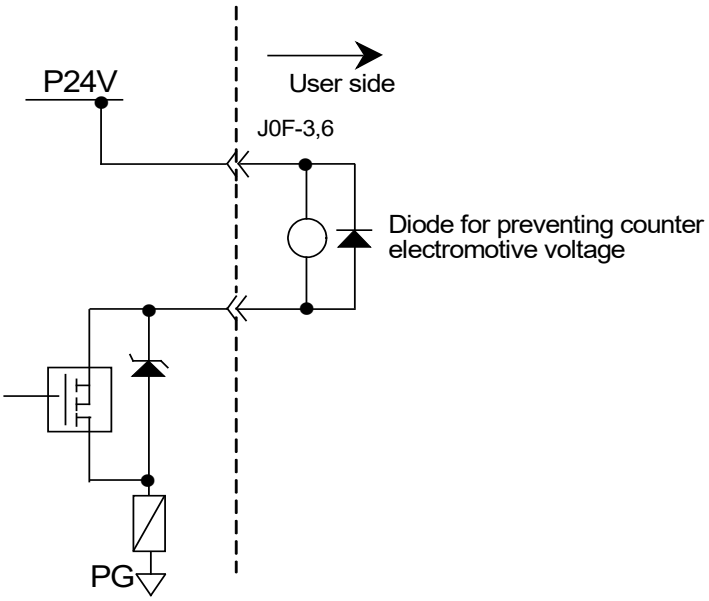
Output specifications:

Rated voltage	:	DC24 V (max. DC30 V)
Rated current	:	1 A (1.2 A or less including other I/Os when the internal power supply is used.)
Leak current	:	100 μA or less

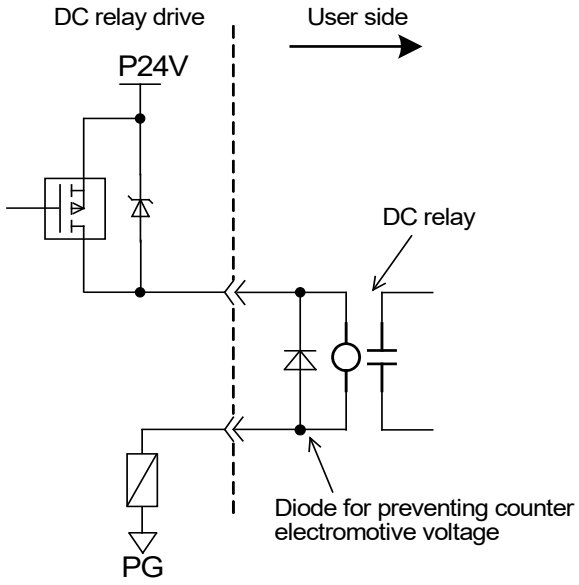
- If the DC24 V power supplied from the robot controller is used, the total current should be 1.2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.
- When a double solenoid is used, HO_1 and HO_2, and HO_3 and HO_4 are used as pairs.

The right figure shows the DC relay circuit when the external power is used.

– Common (X8HN):



+ Common (X8HI):



5.2.2 Tool Wiring

Five (5) input signals are provided for sensors, etc. and four (4) control signals for solenoid valves, etc. A supply power signal of DC24 V is also provided. They are connected to the controller. The wiring arrangement for these cables is shown in Fig. 5.2. The wires are connected to the connectors on the upper side of the arm 2. The user should provide the following connectors to connect the cables.

D-SUB connector (standard)	Type: <Hood> XM2S–1511 (Maker: OMRON) <Body> XM2A–1501 (Maker: OMRON)
Cannon connector (option)	Type: MS3106B–20–29–S (Maker: DDK)
Adaptive cable	Conductive cross section area: 0.2 mm^2 ~ 0.5 mm^2

Each connector and cable are connected by soldering.

! DANGER

- Be sure to use the designated wire. Otherwise, fires or faults may be caused.
- When connecting the connector and wires, make sure not to mistake the terminal arrangement.
- After making the connection, use a tester, etc., to confirm the connection.

When controlling the robot from the sequencer (i.e., programmable ladder controller: PLC), etc. installed separately, remove the motor cover from the base section, remove connectors JOES and JOFS on the rear side, then connect the cables running from the PLC, etc. through the cable inlets provided on the motor cover. (See Fig. 5.2.) For ahead of the JOES and JOFS connectors, the user should prepare the following plug connectors and connect the cables. The current is 1 A or less per cable.

Type of connector:	JOES	SMP–07V–BC (Maker: J.S.T. Mfg.)
	JOFS	SMP–06V–BC (Maker: J.S.T. Mfg.)
Type of contact:	BHF–001T–0.8SS (Maker: J.S.T. Mfg.)	
Adaptive cable	Conductive cross section area: 0.2 mm^2 ~ 0.3 mm^2	

Opposite connector type

Type of connector:	JOEP	SMR–07V–B (Maker: J.S.T. Mfg.)
	JOFP	SMR–06V–B (Maker: J.S.T. Mfg.)

Type of contact: BYM–001T–0.6 (Maker: J.S.T. Mfg.)

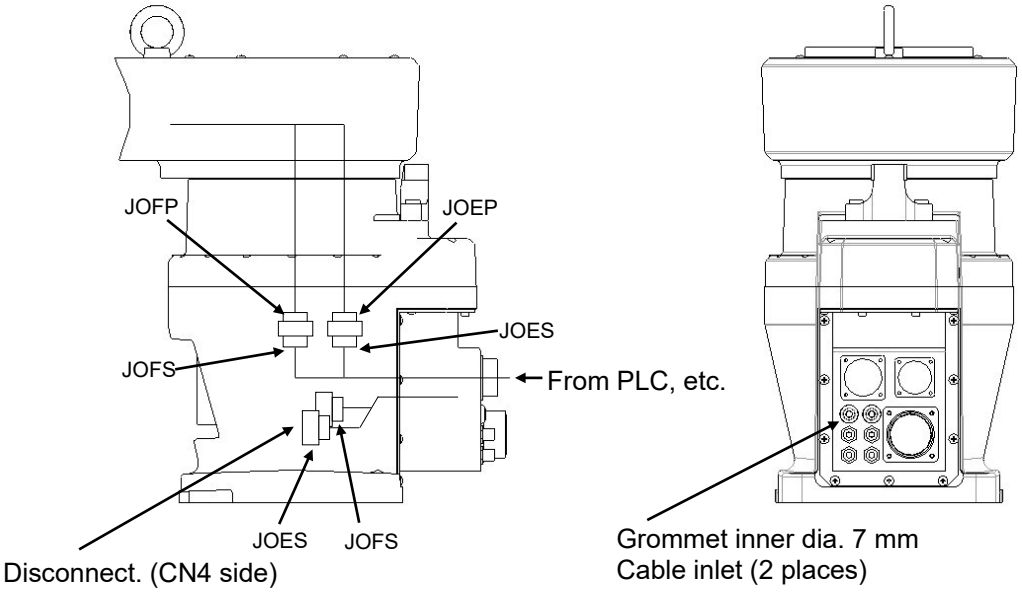
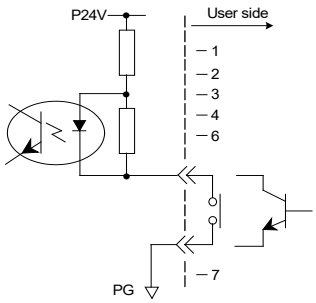
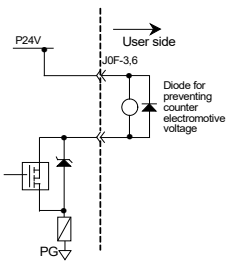
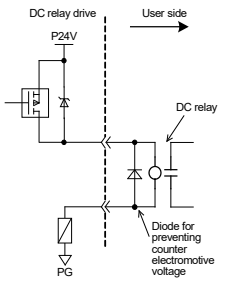


Fig. 5.2 Wiring to PLC, etc.

Input/output signal connector CN0

Pin (D-SUB)	Pin (Cannon)	Signal name		Signal No.	Input/output circuit and example of connections
	A	Not used			
	B	Not used			
	C	Not used			
3	D	GRP1 OPN	Grip 1 open	201	
4	E	GRP1 CLS	Grip 1 close	202	
5	F	GRP2 OPN	Grip 2 open	203	
6	G	GRP2 CLS	Grip 2 close	204	
7	H	WORK	Workpiece	205	
8	J	PG	0 V		Minus common (X8HN) 
	K	Not used			
10	L	GRP1	Grip 1	201	
11	M	$\overline{\text{GRP1}}$	– Grip 1	202	
12	N	P24V	DC24 V power supply		
13	P	GRP2	Grip 2	203	
14	R	$\overline{\text{GRP2}}$	– Grip 2	204	
15	S	P24V	DC24 V power supply		
9	T	Shield			Plus common (X8HI) 

As input signals, no-voltage contacts or transistor open collector inputs are used.

No-voltage contact specifications:

Contact rating: DC24 V, 10 mA or over (circuit current: approx. 7 mA)
Minimum contact current: DC24 V, 1 mA
Contact impedance: 100 Ω or less

Transistor specifications:

Withhold voltage between collector and emitter: 30 V or over
Current between collector and emitter: 10 mA or over (circuit current: approx. 7 mA)
Leak current between collector and emitter: 100 μA or less

By using the DC24 V power of the controller, a relay, solenoid valve, etc., can be driven. When the external power is used, GND of the external power should be common to GND (PG) of the robot controller.

Output specifications:

Rated voltage	:	DC24 V (max. DC30 V)
Rated current	:	1 A
Leak current	:	100 μA or less

- If the DC24 V power supplied from the robot controller is used, the total current should be 2 A or less.
- When the external power is used, the total current should also be 2 A or less.
- When a relay or solenoid valve, etc., is connected, it is necessary to use a surge killer or diode to absorb the surge voltage.

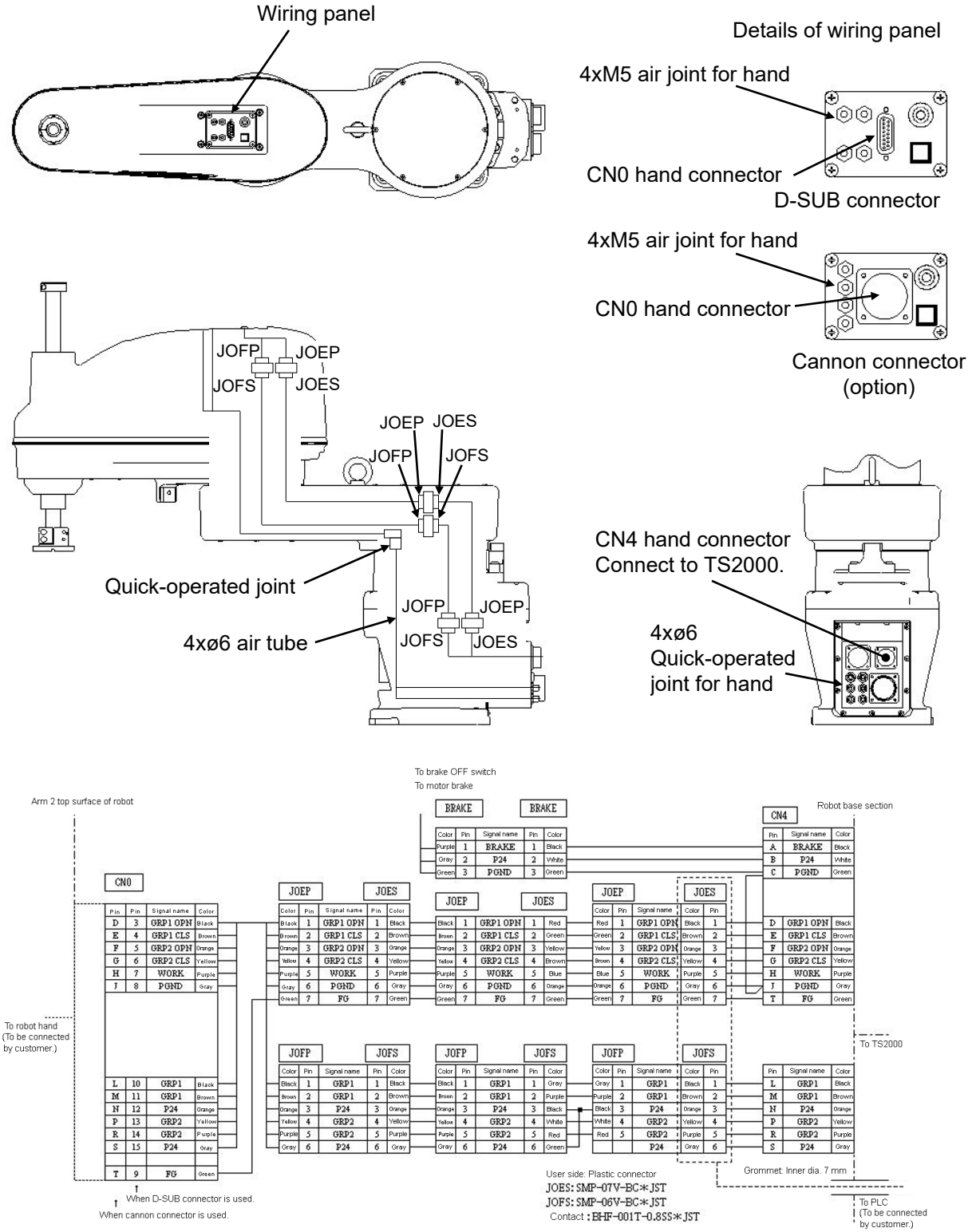
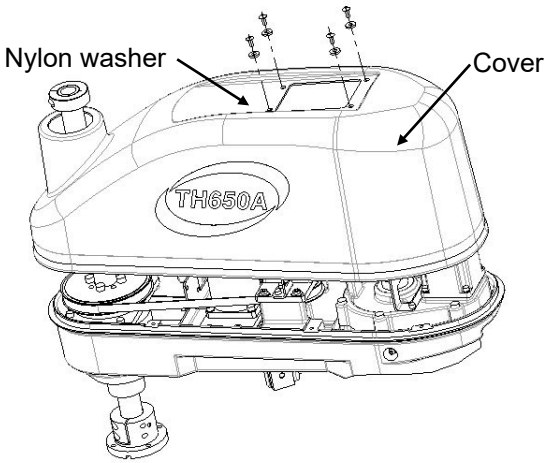


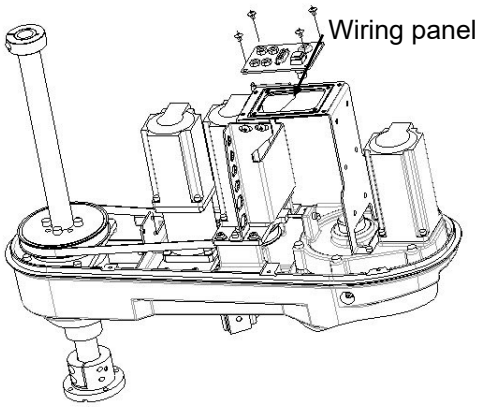
Fig. 5.3 Tool wiring

The cables are connector-connected at the upper side of the arm 2 cover. The wiring panel on the upper side of the same cover can also be mounted on the lower side of the arm 2. The following three (3) manners are available for the tool wiring and piping.

- Wiring and piping to the tool through a hollow hole (18 mm-dia.) on the tool shaft.
- Wiring and piping, using fixed stays.
- Direct wiring and piping by relocating the wiring panel to the lower side.

Fig. 5.4 and Fig. 5.5 show the procedures for relocating the wiring panel and the wiring method, respectively.

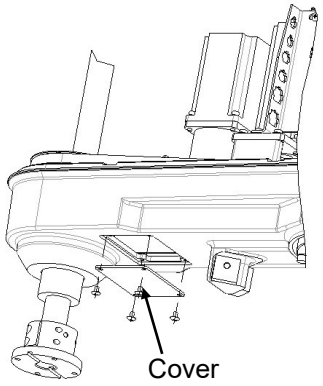




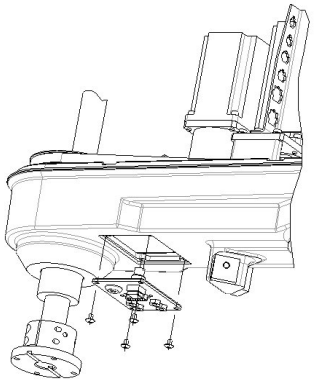
1. Remove the four (4) screws securing the cover, then draw it upward and disconnect.

2. Remove the four (4) screws securing the wiring panel, then dismount the wiring panel. As all cables are connector-connected, disconnect all cables connected there. The gasket for the panel should remain attached to the steel plate securing the cover.

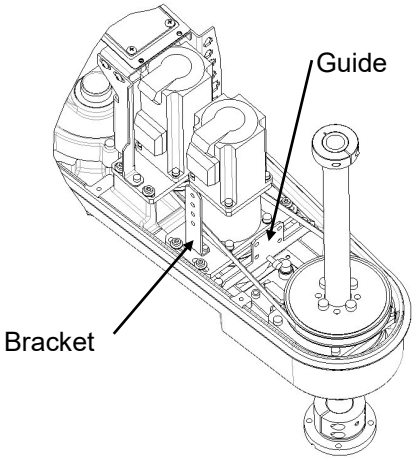
   Connectors to be removed in Step 2:
   JOEP, JOFP, JOES, JOFS
   Brake OFF switch cable, four (4) air tubes



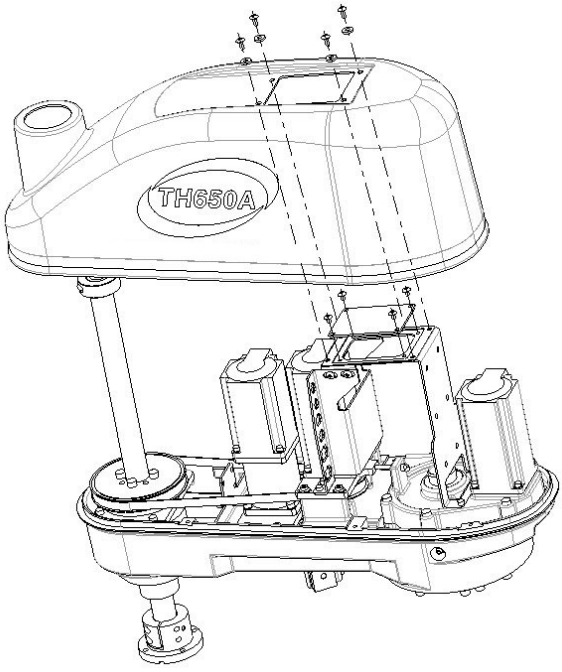
3. Remove the cover at the lower side of the arm 2.



4. Mount the wiring panel which was dismounted in Step 2.



5. The cables connecting the panel with the actuator are fabricated long so that the panel can be mounted on the lower side, and are secured to the bracket with TY-RAP.



6. Finally, secure the cover disconnected in Step 3 to the location where the panel was mounted before, then attach the cover. At this time, be sure to set the nylon washers.
Take careful precautions not to damage the cover.

Cut the secured TY-RAP, using nippers, etc., extend the cables, then connect the connectors coming from the panel. After the connection, secure the cables to the guide, using the TY-RAP so that the cables will not collide with the axis 3 or 4 belt. Secure any loosened cable to the bracket also, using the TYRAP.

Fig. 5.4 Procedures for relocating wiring panel

! CAUTION

- Be sure to connect all connectors properly. Otherwise, the robot may malfunction.
- Take careful precautions not to interfere the cables with the belts. Otherwise, the cables may break.
- When relocating the panel, take utmost care not to damage the belts.

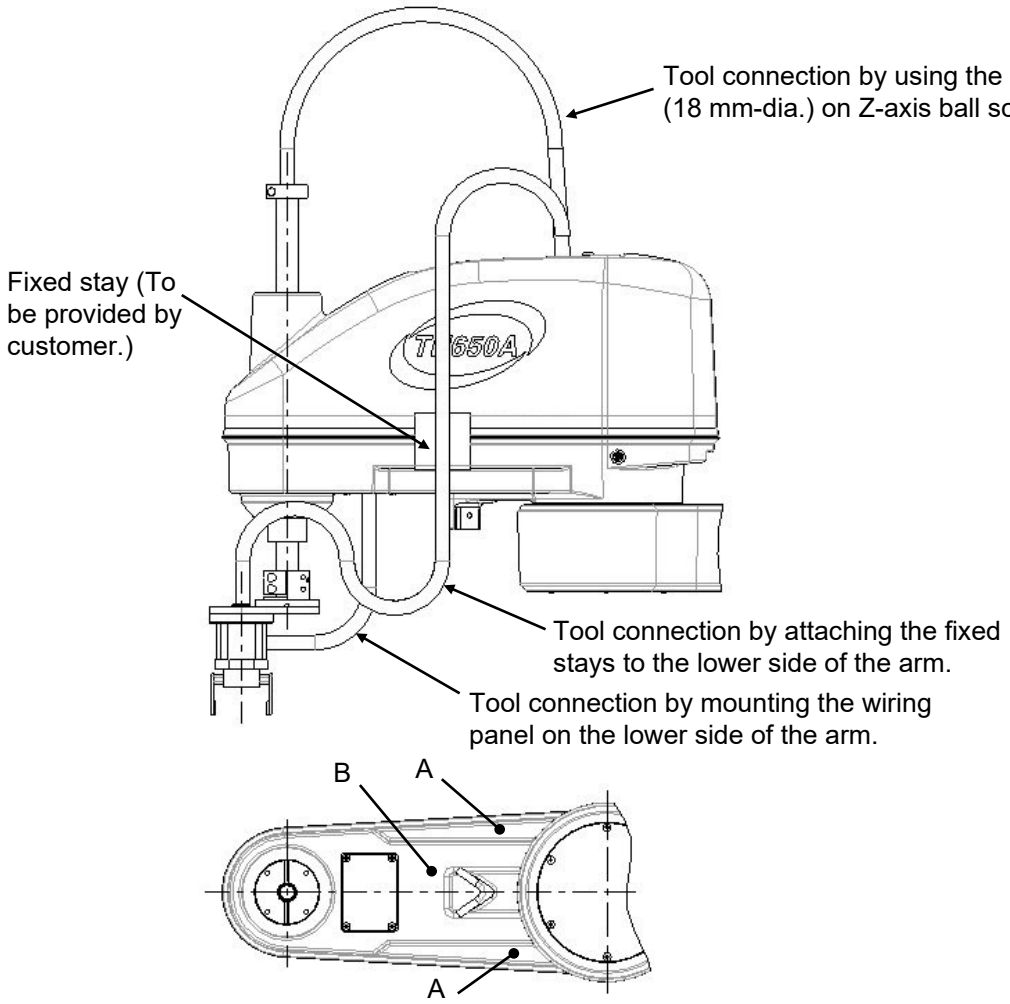


Fig. 5.5 Wiring method

Note: For the tap holes on the lower side of the arm 2, which are used to attach the fixed stays, it is recommended to machine surfaces A in the figure above. If the pre-drilled hole is under 15 mm-deep at this time, it will not go through the top surface.

Recommended tap holes machined: M4 depth 8, pre-drilled hole depth 13 mm

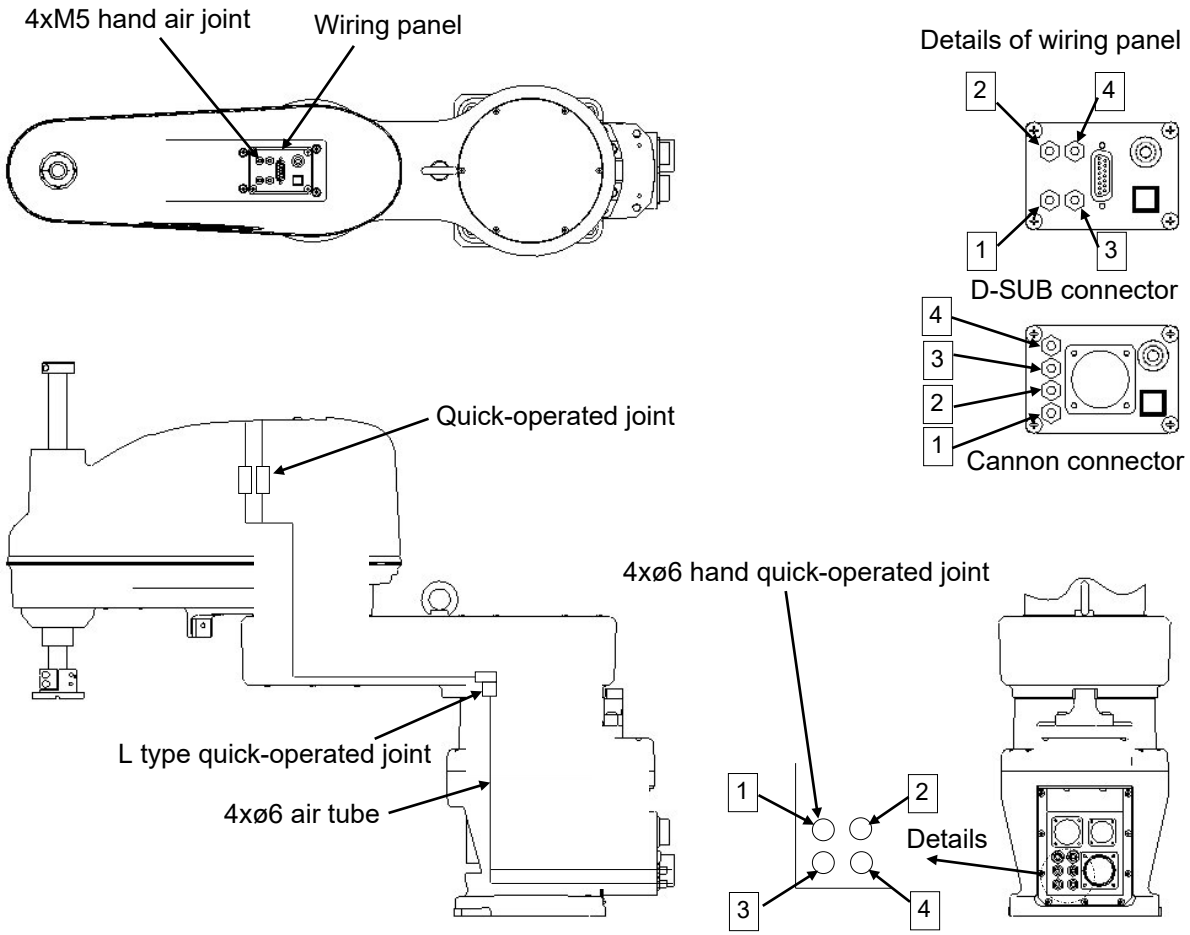
When machining surface B, the hole will go through the top surface (arm 2 interior). When using the robot in a clean environment, machine surfaces A and attach the stays there.

! CAUTION

- Be sure to use a highly flexible robot cable, which should be secured below the arm with a cable clamp, etc. Unless a robot cable is used, the wire may be broken.
- When performing tool wiring and piping, take all necessary measures against breakage due to rub, etc.
- At robot operation, take careful precautions not to exert a load on respective connectors.

5.2.3 Tool Air Piping

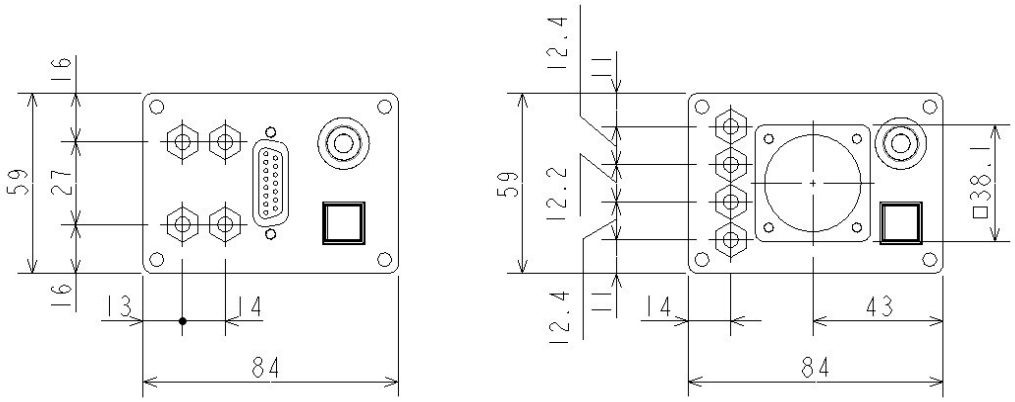
The robot is provided with four (4) air lines for the tool.
The outer diameter of the air pipelines is 6 mm. Fig. 5.6 shows the tool air piping.
The air control unit (oiler, regulator with gage and filter) for the solenoid valves should be provided by the user.



The air tube is identified by the number and color. At piping, make sure that each tube is connected properly, referring to the below-mentioned.

1 : Red 2 : White 3 : Blue 4 : Yellow

Air joint pitches of the panel



D-SUB connector Cannon connector (option)

Fig. 5.6 Tool air piping

5.3 Permissible Load Conditions and Program Setting

This paragraph describes the permissible load conditions of the robot and how to set up the program according to the load.

5.3.1 Permissible Load Conditions

The robot load conditions are defined by the tool mass, moment of inertia and offset value of tool gravity center from the center of the tool shaft, as shown in Fig. 5.7.
The permissible load conditions are shown in Table 5.1.

Table 5.1 Permissible load conditions

Conditions	Permissible values
Mass	Max. 10 kg
Load inertia	Max. 0.10 kg·m^2
Offset value of load gravity center	Max. 100 mm

! CAUTION

- NEVER operate the robot under the load conditions exceeding the permissible values. Otherwise, the robot life and safety cannot be guaranteed.

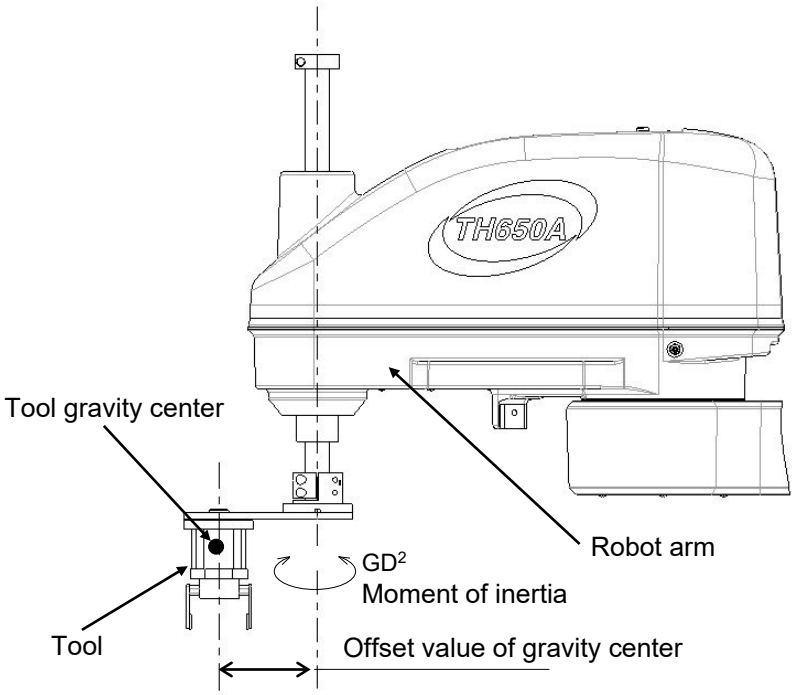


Fig. 5.7 Robot tool

5.3.2 Load Conditions and Program Setting

This robot can automatically change the maximum speed, acceleration/deceleration and servo gain by using the PAYLOAD command in the program according to the load conditions.

Be sure to use the PAYLOAD command.

The specific method for using this function is explained below.

a) PAYLOAD command format

The PAYLOAD command format is written as shown below if the tool mass is M kg and the gravity center offset is L mm.

PAYLOAD = {M, L}

M : Load mass (unit: kg)

L : Offset value of gravity center (unit: mm)

The PAYLOAD command has the following functions.

- The maximum speed and acceleration/deceleration of each robot axis are automatically changed according to the set load conditions.

- The servo gain of each robot axis is automatically changed according to the set load conditions.

b) Program examples

Basic program examples using the PAYLOAD command are shown below. For further information, see the Robot Language Manual.

(Program example 1)

The robot is moved under the load conditions of 5 kg mass and 100 mm gravity center offset.

```
PROGRAM SAMPLE
  SPEED=100
  PAYLOAD={5,100}
  MOVE P1
  MOVE P2
  STOP
END
```

(Program example 2)

When the hand mass is 3 kg and the gravity center offset is 30 mm, and the mass is 5 kg and gravity center offset is 50 mm when the workpiece is grasped.
Pick-and-place operation is executed under the above conditions.

```
PROGRAM SAMPLE
  PAYLOAD={3,30}
  ACCUR=COARSE
  ENABLE NOWAIT
  RESET DOUT
  MOVE P0
  DOUT(1)
  WAIT DIN(1)
LOOP:
  MOVE P1+POINT(0,0,100)
  IF DIN(-1)THEN GOTO FIN
  MOVE P1
  WAIT MOTION>=100
  DOUT(213)
  DELAY 1
  PAYLOAD={5,50}
  MOVE P1+POINT(0,0,100)
  MOVE P2+POINT(0,0,100)
  MOVE P2
  WAIT MOTION>=100
  DOUT(-213)
```

```
    DELAY 1
    PAYLOAD={3,30}
    MOVE P2+POINT(0,0,100)
    GOTO LOOP
FIN:
    MOVE P0
    DOUT(1)
    STOP
END
```

c) Setting of PAYLOAD command

In the default state, or when the PAYLOAD command is not used, the maximum speed and acceleration/deceleration are set to 100 % and the servo gain is set to the value under the minimum load. (See Para. 5.3.3.)

! CAUTION

- Be sure to use the PAYLOAD command.
- Unless the PAYLOAD command is used, the robot will vibrate or overshoot, resulting in malfunction or shortening of the life of the mechanisms. In the worst case, the mechanism will be damaged.
- Even when the PAYLOAD command is used, regulate the speed by using the SPEED or DECEL command while confirming the workpiece behavior subject to handling.

! CAUTION

- The load moment of inertia should be within the tolerances given in Table 5.1.
- Even if there is no offset of load gravity center, if the moment of inertia is large, the robot may vibrate. When this happens, figure out the virtual gravity center offset (L mm) from the following equation, using the moment of inertia (J kg·m^2) and mass (M kg).

  $L = \sqrt{(J \times 10^6 / M)}$

  Then, designate the following command.

  PAYLOAD = {M, L}

! CAUTION

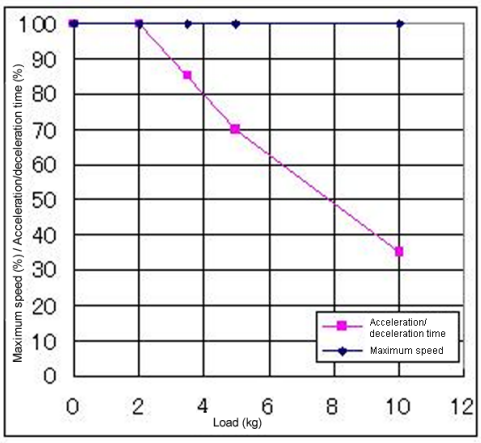
- When guiding manually, the robot may vibrate if the load mass or gravity center offset is large. This is because the servo gain is not appropriate. When this happens, perform the following operation while setting the load conditions in the test run mode.

  DO PAYLOAD = {M, L} M: Mass [kg] L: Offset [mm]
  EXE

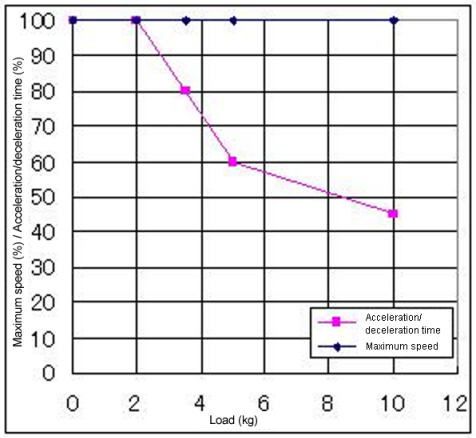
  The servo gain is changed then to the value which meets the load conditions.

5.3.3 Setting Maximum Speed and Robot Acceleration/Deceleration for Load Conditions

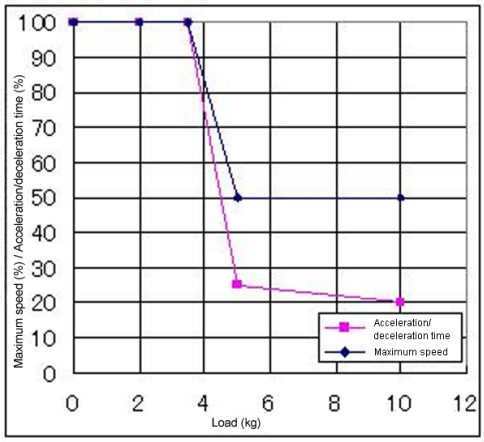
When the PAYLOAD command is used, the maximum speed and acceleration/deceleration of the robot are automatically changed according to the load conditions.
The maximum speed and acceleration/deceleration change with the load mass, as shown in Fig. 5.8.



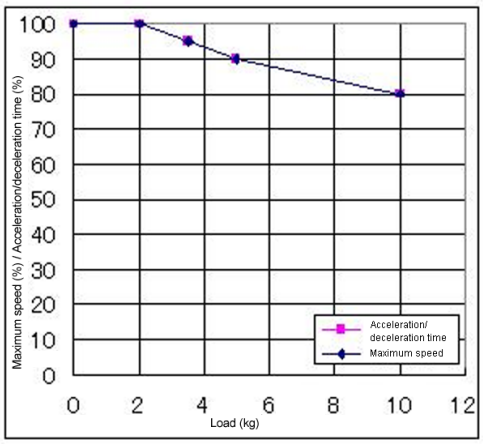
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 1)



Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 2)



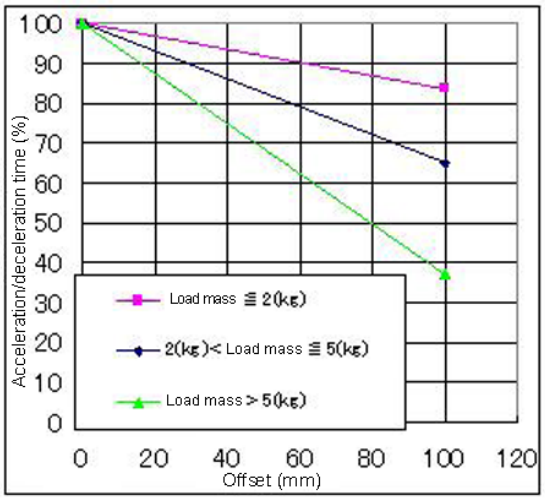
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 3)



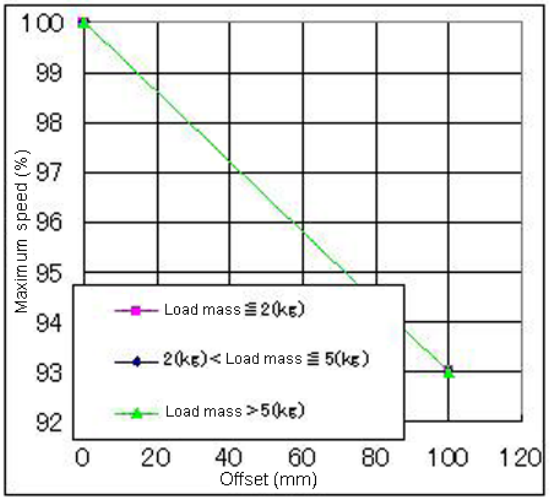
Setting of maximum speed and acceleration/deceleration in relation to load mass (Axis 4)

Fig. 5.8 Setting of maximum speed and acceleration/deceleration in relation to load mass

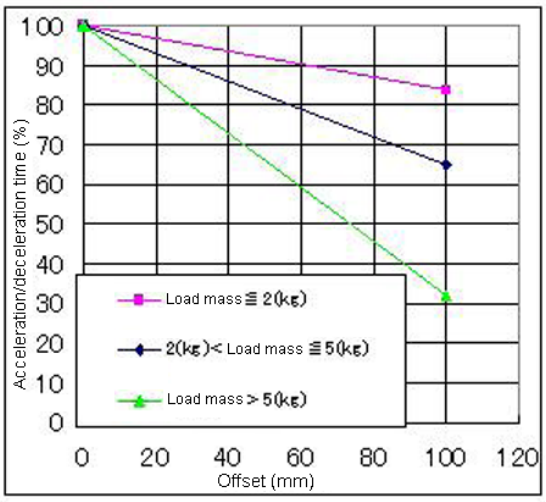
Additionally, if there is an offset of load gravity center, the maximum speed and acceleration/deceleration change as shown in Fig. 5.9 and Fig. 5.10. For the axis 3, however, no restriction is imposed on the offset.



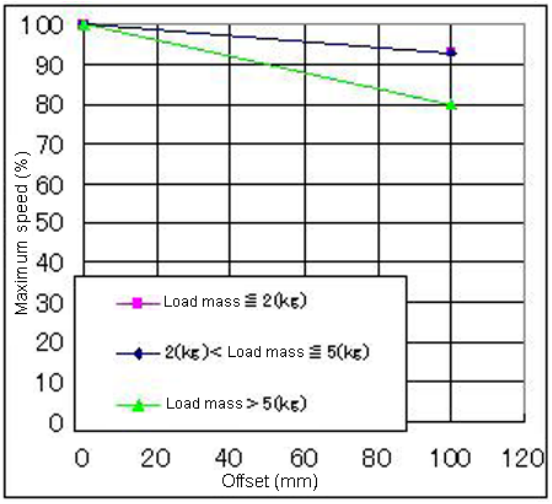
Setting of acceleration/deceleration in relation to offset (Axis 1)



Setting of maximum speed in relation to offset (Axis 1)

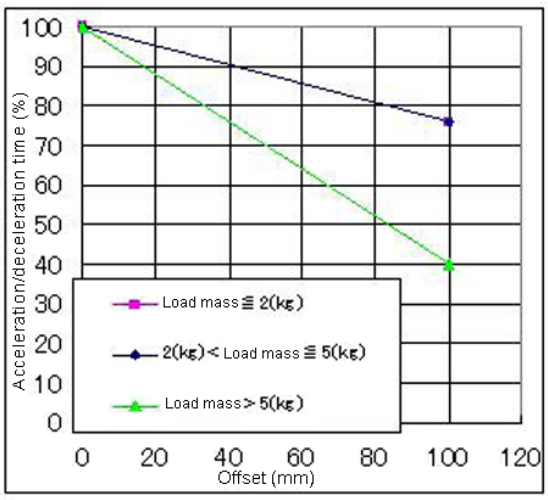


Setting of acceleration/deceleration in relation to offset (Axis 2)

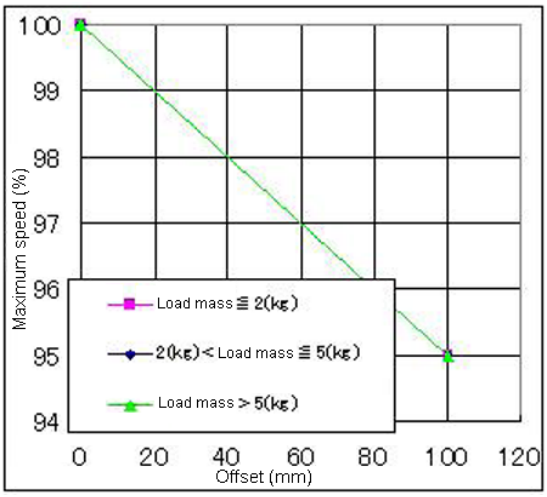


Setting of maximum speed in relation to offset (Axis 2)

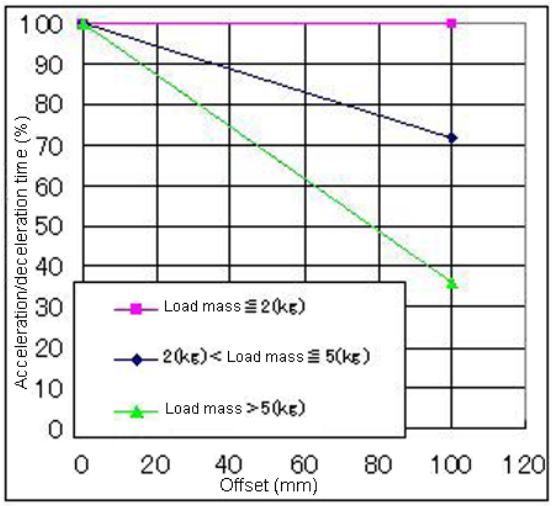
Fig. 5.9 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 1 and 2)



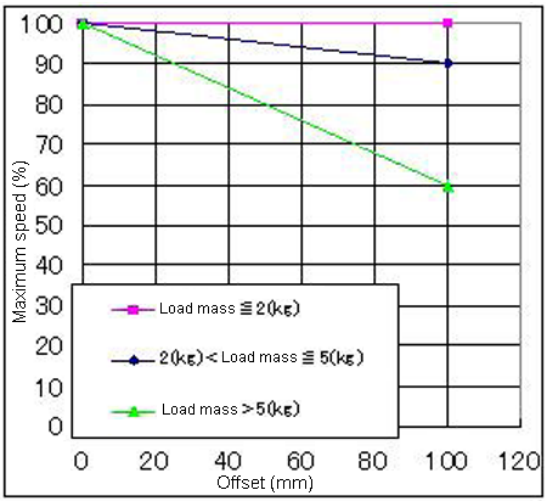
Setting of acceleration/deceleration in relation to offset (Axis 3)



Setting of maximum speed in relation to offset (Axis 3)



Setting of acceleration/deceleration in relation to offset (Axis 4)



Setting of maximum speed in relation to offset (Axis 4)

Fig. 5.10 Setting of maximum speed and acceleration/deceleration in relation to gravity center offset (Axes 3 and 4)